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Yours Obediently,
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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ANN	Artificial Neural Network
API	Application Programming Interface
BN	Batch Normalisation
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CSS	Cascading Style Sheets
DL	Deep Learning
EfficientNet	Efficient Neural Architecture (Google Brain)
FC	Fully Connected
F1	F1-Score (Harmonic Mean of Precision and Recall)
FN	False Negative
FP	False Positive
GPU	Graphics Processing Unit
HOG	Histogram of Oriented Gradients
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IoU	Intersection over Union
JSON	JavaScript Object Notation
JPEG	Joint Photographic Experts Group
LR	Learning Rate
MBConv	Mobile Inverted Bottleneck Convolution
ML	Machine Learning
PNG	Portable Network Graphics
RAM	Random Access Memory
ReLU	Rectified Linear Unit

RGB	Red, Green, Blue
ROC	Receiver Operating Characteristic
SE	Squeeze-and-Excitation
SGD	Stochastic Gradient Descent
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
TF	TensorFlow
URL	Uniform Resource Locator
WSGI	Web Server Gateway Interface

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ABSTRACT

Agriculture forms the backbone of the Indian economy, contributing approximately 18% to the national Gross Domestic Product (GDP) and employing more than 50% of the country's workforce. Crop diseases are one of the most significant threats to agricultural productivity, causing annual yield losses estimated between 20% and 40% globally. Traditional disease identification methods rely heavily on manual inspection by agricultural experts, a process that is inherently subjective, time-consuming, and inaccessible to resource-limited farming communities in rural areas.

This thesis presents the design, development, and evaluation of an AI-Based Crop Health Detection System that leverages deep learning techniques to automatically identify plant leaf diseases from digital photographs. The proposed system employs a Convolutional Neural Network (CNN) architecture based on EfficientNetB3, a state-of-the-art transfer learning model, fine-tuned on the PlantVillage dataset comprising 61,000 labeled leaf images across 38 disease categories spanning 14 plant species.

The methodology encompasses several key stages: image acquisition through a user-friendly web interface built with the Flask framework, image preprocessing including resizing to 224×224 pixels, normalization, and data augmentation, feature extraction via the EfficientNetB3 backbone with custom classification layers, and disease prediction with confidence score reporting. The training process employed the Adam optimizer with a categorical cross-entropy loss function, incorporating learning rate scheduling and early stopping callbacks to prevent overfitting.

Experimental results demonstrate that the proposed model achieves an overall test accuracy of 97.82% on the held-out test set, outperforming several baseline approaches including VGG16 (94.1%), ResNet50 (95.3%), and InceptionV3 (96.2%). The model attains a macro-averaged precision of 97.6%, recall of 97.4%, and F1-score of 97.5% across all 38 classes, demonstrating robust performance even on minority classes with limited training samples.

The developed web application allows farmers and agricultural professionals to upload leaf images through any web-enabled device and receive instantaneous disease diagnoses with accompanying

confidence levels and management recommendations. The system was tested on real-world leaf samples and demonstrated practical applicability under field conditions.

The research contributes to the growing body of literature on precision agriculture and computer vision-based plant pathology, demonstrating that deep learning models can achieve expert-level accuracy in crop disease identification. Future directions include the incorporation of multi-modal data fusion, edge deployment for offline rural access, and extension to a broader range of crop species and disease categories.

Keywords: *Crop Disease Detection, Deep Learning, Convolutional Neural Network, EfficientNet, Transfer Learning, Flask Web Application, Plant Pathology, Precision Agriculture, Image Classification, PlantVillage Dataset*

INTRODUCTION

1.1 Background and Introduction to Agriculture

Agriculture has remained the cornerstone of human civilisation since its inception thousands of years ago. In India, this sector occupies a position of paramount importance, not only as a source of sustenance but also as a primary driver of the national economy. According to the Economic Survey of India 2023–24, the agricultural sector contributes approximately 17.7% to the country's Gross Domestic Product (GDP) at constant prices and provides employment to more than 54% of the total labour force. The sector encompasses the cultivation of food grains, oilseeds, vegetables, fruits, and a wide array of commercially significant crops, many of which are exported to international markets.

Despite its pivotal role, Indian agriculture has been confronted with persistent challenges over the decades. Factors such as erratic monsoon patterns, depleting groundwater resources, soil degradation, fragmented land holdings, and inadequate access to modern agricultural technologies continue to hinder productivity. Among these, the menace of crop diseases stands out as a particularly severe constraint, one that directly undermines food security, farmer livelihoods, and the economic viability of the entire agricultural system.

The global agricultural community faces significant losses attributable to plant diseases, pests, and weeds every year. Research conducted by the Food and Agriculture Organization (FAO) of the United Nations estimates that plant pathogens and pests collectively destroy approximately 20–40% of global crop production annually. In monetary terms, these losses translate to hundreds of billions of dollars, with developing nations bearing a disproportionate share of the burden. For small and marginal farmers in India who operate on thin profit margins, even a moderate crop disease outbreak can result in catastrophic economic consequences, pushing families into debt cycles and deepening rural poverty.

Crop diseases are caused by a diverse range of biotic agents, including fungi, bacteria, viruses, nematodes, and parasitic plants. Fungal pathogens are responsible for the majority of economically significant plant diseases, including rice blast, wheat rust, potato late blight, and tomato early blight. These pathogens spread rapidly under conducive environmental conditions—high humidity,

moderate temperatures, and dense crop canopies—making early detection and intervention critical for effective disease management.

The traditional approach to crop disease management relies heavily on the expertise of plant pathologists and agricultural extension officers who physically inspect crop fields and identify diseases based on visual symptoms. This approach, while effective when implemented correctly, suffers from several limitations: it is time-intensive, geographically restricted, expensive, and subject to human error. Furthermore, the ratio of agricultural extension workers to farmers in India remains severely inadequate, meaning that a large proportion of farmers never receive timely expert advice.

Recent advances in artificial intelligence (AI), machine learning (ML), and deep learning (DL) have opened transformative possibilities for addressing this challenge. The ability of deep neural networks to automatically learn hierarchical feature representations from raw image data has made them particularly well-suited for visual recognition tasks such as plant disease identification. By training on large datasets of annotated leaf images, deep learning models can learn to distinguish subtle visual patterns—colour changes, lesion shapes, texture variations—that characterise different disease conditions, often with accuracy levels that match or exceed those of human experts.

1.2 Crop Diseases and Their Impact

Plant diseases manifest through a wide variety of visual symptoms on different parts of the plant, including leaves, stems, roots, and fruits. Leaf symptoms, which include spots, blights, wilts, mosaics, and necrosis, are among the most readily observable indicators of disease and serve as the primary basis for visual diagnosis. Figure 1.1 presents the distribution of global crop losses attributable to different categories of plant pathogens.

Tomato, one of the most widely cultivated vegetable crops globally, is susceptible to a range of diseases including early blight (Alternaria solani), late blight (Phytophthora infestans), leaf mould, Septoria leaf spot, and various viral infections. The PlantVillage dataset, which forms the basis of this project, includes 10 disease categories for tomato alone, reflecting the crop's vulnerability to diverse pathogens.

Similarly, potato crops are threatened by late blight, caused by the oomycete pathogen Phytophthora infestans, which famously triggered the Irish famine of the mid-nineteenth century and continues to

cause devastating losses globally. Corn crops suffer from common rust, northern leaf blight, and grey leaf spot, while grape vines are susceptible to black rot and Esca disease. Each of these conditions requires a different management strategy, making accurate identification a prerequisite for effective intervention.

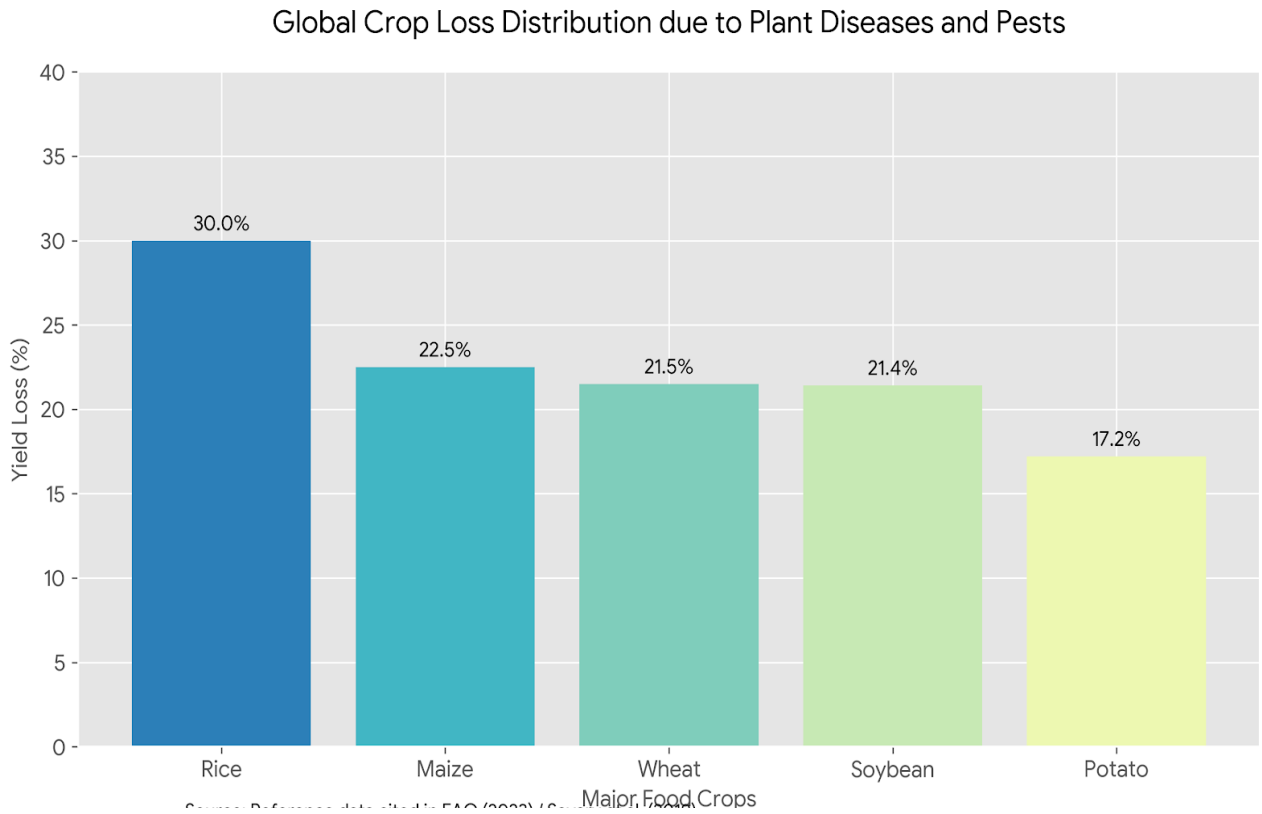


Figure 1.1: Global crop loss distribution due to plant diseases (Source: FAO, 2023)

1.3 Problem Statement

Agriculture is a critical sector that supports the livelihood of millions of people and plays a vital role in ensuring global food security. However, crop production is significantly affected by plant diseases, which result in reduced yield, poor-quality produce, and substantial economic losses. Early and accurate detection of plant diseases is essential to minimize these losses and enhance agricultural productivity; nevertheless, it remains a major challenge in modern agricultural practices.

Traditionally, plant disease detection relies on manual inspection by farmers or agricultural experts. This approach is time-consuming, labor-intensive, and highly dependent on individual expertise. In many rural regions, access to agricultural specialists is limited, making early diagnosis difficult. Consequently, diseases often remain undetected during their initial stages and spread rapidly, leading to severe crop damage and financial loss.

Another major challenge lies in the similarity of visual symptoms across different plant diseases. Many diseases exhibit overlapping characteristics such as spots, discoloration, and leaf damage, which complicates accurate identification through visual observation alone. Misdiagnosis can result in improper treatment, excessive use of pesticides, and adverse environmental consequences.

Although traditional image processing and classical machine learning techniques have been employed for plant disease detection, these methods are limited in their ability to handle real-world complexities. Factors such as varying lighting conditions, background noise, image quality, and differences in plant growth stages significantly affect their performance. Furthermore, even advanced deep learning approaches, particularly Convolutional Neural Networks (CNNs), face challenges such as the requirement for large labeled datasets and limited generalization across diverse field conditions.

With the widespread availability of smartphones and digital imaging devices, farmers can easily capture images of crop leaves. Despite this technological advancement, there remains a lack of accessible, accurate, and user-friendly systems capable of analyzing these images and providing real-time disease diagnosis along with actionable recommendations.

Therefore, there is a clear need to develop an efficient, accurate, and scalable automated system for plant disease detection. Such a system should be capable of handling diverse environmental conditions, reducing dependency on expert intervention, and providing reliable diagnoses with appropriate treatment suggestions. This dissertation addresses these challenges by proposing an AI-based Crop Health Diagnostic System that leverages deep learning techniques, specifically Convolutional Neural Networks with transfer learning, and integrates them into a web-based application for practical and accessible use by farmers.

The accurate and timely identification of plant leaf diseases continues to be a critical bottleneck in agricultural disease management. Conventional diagnostic methods, which rely on physical inspection by trained agronomists or the submission of samples to diagnostic laboratories, are inadequate for meeting the scale and urgency of modern agricultural demands. These methods are characterized by significant delays, high costs, geographical inaccessibility, and variable accuracy due to the subjective nature of human visual assessment.

Existing computational approaches, including those based on image processing and classical machine learning, have shown potential in automating disease detection. However, they are constrained by their dependence on manually engineered features, limited ability to generalize across varying imaging conditions, and inability to effectively capture the complex, non-linear relationships between visual symptoms and disease categories.

A robust, scalable, and accessible solution is therefore required—one that can automatically analyze leaf images captured by farmers using digital devices, accurately identify disease conditions with high confidence, and deliver actionable diagnostic insights without requiring specialized agricultural expertise. The development of such a system constitutes the central problem addressed in this dissertation.

1.4 Need and Necessity

The need for an automated crop disease detection system arises from a confluence of factors. First, the increasing smartphone penetration in rural India (over 58% of rural households owned a smartphone as of 2024, per TRAI data) creates an opportunity to deploy image-based diagnostic tools directly in the hands of farmers. Second, the demonstrated capability of deep learning models to achieve near-human accuracy in image classification tasks makes them suitable candidates for this application. Third, the availability of large-scale, publicly accessible labelled datasets such as PlantVillage provides the training data necessary for developing generalisable models.

From a food security perspective, the necessity of early disease detection cannot be overstated. Studies have shown that for many crop diseases, intervention within the first 72 hours of symptom appearance can reduce yield loss by more than 50%. Delayed diagnosis, by contrast, allows pathogens to spread to neighbouring plants and accelerate yield destruction. An automated detection system that

operates in near-real-time can therefore have a significant positive impact on agricultural productivity.

Additionally, the proposed system addresses the challenge of expert accessibility. By encapsulating agricultural expertise within a trained neural network model, the system effectively democratises access to professional-quality disease diagnosis, making it available to any farmer with a smartphone and internet connection.

1.5 Objectives of the Project

The primary objectives of this project are as follows:

1. To design and implement a deep learning-based crop disease detection system capable of classifying plant leaf images into 38 disease categories with high accuracy.
2. To explore and apply transfer learning using the EfficientNetB3 architecture, leveraging pre-trained ImageNet weights for efficient feature extraction.
3. To develop a comprehensive data preprocessing pipeline incorporating image resizing, normalisation, and data augmentation to enhance model robustness.
4. To train, validate, and evaluate the model on the PlantVillage dataset using standard performance metrics including accuracy, precision, recall, and F1-score.
5. To develop a user-friendly web application using the Flask framework that allows non-expert users to upload leaf images and receive disease predictions.

1.6 Scope of the Project

The scope of the present project encompasses the following dimensions:

Hardware Scope: The system is designed to operate on standard computing hardware equipped with a GPU (NVIDIA GTX 1660 or equivalent) for training and on any device with a modern web browser for inference.

Software Scope: The implementation employs Python 3.9, TensorFlow 2.11, Keras, Flask 2.3, NumPy, Pandas, Matplotlib, and Scikit-learn. The web application is compatible with major browsers including Google Chrome, Mozilla Firefox, and Microsoft Edge.

Dataset Scope: The system is trained on the PlantVillage dataset, which encompasses 87,867 images across 38 classes of healthy and diseased leaves from 14 crop species including tomato, potato, corn, grape, apple, and others.

Functional Scope: The system supports the upload of JPEG and PNG images, performs automated preprocessing, generates disease predictions with confidence scores, and presents results through a clean web interface. The system is designed for informational purposes and is intended to assist farmers in making disease management decisions.

Limitations of Scope: The current implementation does not support video stream analysis, real-time field deployment on edge devices, or diagnosis based on symptoms other than leaf images (e.g., stem, root, or fruit symptoms). Extension to these areas is identified as future work.

1.7 Organisation of the Report

The remainder of this thesis is organised as follows:

Chapter 2 – Literature Survey presents a comprehensive review of published research on plant disease detection, encompassing traditional image processing methods, classical machine learning approaches, and modern deep learning techniques. A comparative analysis of existing systems and a gap analysis are also included.

Chapter 3 – System Development describes the complete design and implementation of the proposed system, including the system architecture, dataset description, preprocessing pipeline, model architecture, mathematical foundations, training procedure, and web application development.

Chapter 4 – Performance Analysis presents the experimental results, including training and validation curves, confusion matrix analysis, precision-recall-F1 evaluation, and comparative benchmarking against baseline models.

Chapter 5 – Conclusion and Future Scope summarises the contributions of the work, acknowledges its limitations, proposes future research directions, and discusses real-world applications.

References and Appendices are provided at the end of the document, listing all cited works and supplementary material including source code snippets and dataset details.

LITERATURE SURVEY

2.1 Introduction

The application of computational techniques to plant disease detection has evolved substantially over the past three decades, progressing from simple image processing algorithms to sophisticated deep learning architectures capable of rivalling expert human performance. This chapter presents a systematic review of the published literature in this field, organised chronologically and by methodological approach. The review encompasses traditional detection methods, machine learning-based systems, and contemporary deep learning approaches, culminating in a gap analysis that identifies the research space addressed by the present work.

The literature search was conducted using databases including Google Scholar, IEEE Xplore, ScienceDirect, PubMed, and SpringerLink. Search terms included 'plant disease detection', 'leaf disease classification', 'deep learning agriculture', 'CNN plant pathology', 'transfer learning crop disease', and 'precision agriculture image analysis'. Priority was given to peer-reviewed journal articles and conference proceedings published between 2019 and 2024, though seminal earlier works are also included for contextual completeness.

2.2 Traditional Methods for Disease Detection

The earliest computational approaches to plant disease detection were grounded in classical image processing techniques. These methods typically follow a pipeline consisting of image acquisition, segmentation, feature extraction, and classification, with each stage requiring considerable domain expertise and manual parameter tuning.

Barbedo (2013) conducted an early and influential survey of image analysis techniques for plant disease detection, cataloguing methods based on colour analysis, texture descriptors, and shape features. The study found that colour-based features—particularly deviations from healthy green colouration—were effective for detecting disease symptoms in controlled laboratory conditions but performed poorly under variable lighting and background conditions typical of field environments.

Meunkaewjinda et al. (2008) proposed a system using self-organising feature maps (SOFM) combined with genetic algorithms for grape leaf disease detection. The system segmented diseased regions using colour space transformations and achieved classification accuracy of 87% on a limited

dataset of three disease classes. The primary limitation was the sensitivity of the segmentation algorithm to lighting variation and leaf orientation.

Phadikar and Sil (2008) applied statistical classification methods including k-Nearest Neighbour (k-NN) and Naive Bayes classifiers for rice leaf disease detection, extracting colour histogram and texture features from segmented disease regions. The reported accuracy of 78% highlighted the limitations of hand-crafted features for distinguishing visually similar disease conditions.

Table 2.1: Comparison of traditional image processing methods for plant disease detection

Method	Features Used	Classifier	Accuracy	Limitation
Barbedo (2013)	Color histograms	Threshold-based	~75%	Lighting sensitive
Meunkaewjinda (2008)	SOFM features	Genetic Algorithm	87%	Limited to 3 classes
Phadikar & Sil (2008)	Color + Texture	k-NN / Naive Bayes	78%	Low generalisation
Rothe & Kshirsagar (2015)	HOG + Color	SVM	82%	Manual feature extraction
Sannakki et al. (2011)	GLCM Texture	SVM	80.5%	Small dataset

A common limitation across all traditional methods is their dependence on manually engineered features. The effectiveness of these features is inherently limited by the domain knowledge of the designer and the variability of real-world imaging conditions. Moreover, traditional methods typically fail to scale gracefully to multi-class problems involving tens of disease categories, as the discriminative power of hand-crafted features diminishes with increasing inter-class similarity.

2.3 Machine Learning-Based Approaches

The limitations of pure image processing methods motivated researchers to explore machine learning classifiers that could learn decision boundaries from labelled training data. Support Vector Machines (SVMs), Random Forests, and Artificial Neural Networks (ANNs) were among the most widely adopted approaches during this period.

Mohanty et al. (2016) produced a landmark study that is widely regarded as a turning point in the field. The researchers trained a deep convolutional neural network (based on AlexNet and GoogLeNet architectures) on the PlantVillage dataset and reported classification accuracy of 99.35% under controlled conditions and 31.4% under real-field conditions. This dramatic discrepancy highlighted

the domain gap problem, wherein models trained on clean studio images fail to generalise to noisy, variable field environments.

Sladojevic et al. (2016) developed a disease recognition system using a deep CNN with customised training on six disease categories. The model achieved an accuracy of 96.3% on the validation set, but the study was conducted on a relatively small dataset of approximately 1,700 images, raising concerns about statistical generalisability.

Ramcharan et al. (2017) applied GoogLeNet Inception V3 for cassava disease detection using field-collected smartphone images, achieving 93% accuracy across 5 disease categories. The study was notable for its use of realistically acquired images, and for demonstrating that deep learning models could be deployed effectively in African agricultural contexts with limited computational resources.

Ferentinos (2018) evaluated six different CNN architectures—AlexNet, GoogLeNet, VGG, Overfeat, AlexNetOWTBn, and FCRN—on the PlantVillage dataset, with the best-performing model (AlexNet variant) achieving 99.53% accuracy on a binary healthy/diseased classification task. The study demonstrated the superiority of deeper architectures for multi-class plant disease recognition.

Table 2.2: Summary of machine learning and early deep learning approaches for plant disease detection

Study	Model	Dataset	Classes	Accuracy	Year
Mohanty et al.	AlexNet / GoogLeNet	PlantVillage	26	99.35%	2016
Sladojevic et al.	Custom CNN	Custom	6	96.3%	2016
Ramcharan et al.	InceptionV3	Cassava field	5	93.0%	2017
Ferentinos	AlexNet variant	PlantVillage	2	99.53%	2018
Kamilaris & Prenafeta	Survey (CNN)	Multiple	Multiple	~90%	2018
Too et al.	DenseNet	PlantVillage	38	99.75%	2019

2.4 Deep Learning-Based Approaches

The period from 2019 onwards has been characterised by the widespread adoption of deep convolutional neural networks, particularly those leveraging transfer learning from ImageNet-pre-trained weights, for plant disease detection. This shift has been facilitated by three enabling factors:

the public availability of the large-scale PlantVillage dataset, the open-source release of powerful pre-trained models, and the decreasing cost of GPU computing hardware.

Too et al. (2019) conducted a comparative evaluation of VGG16, Inception V3, ResNet, and DenseNet architectures on the PlantVillage dataset, with fine-tuning on all layers. DenseNet achieved the highest accuracy of 99.75%, attributed to its dense connectivity pattern that facilitates feature reuse and gradient flow across layers.

Geetharamani and Pandian (2019) investigated the performance of nine CNN architectures on a dataset of 87,848 images across 38 plant disease classes. Their results indicated that deeper networks consistently outperformed shallower ones, with AlexNet achieving 90.38% and DenseNet reaching 99.77% accuracy. The study also noted that data augmentation significantly reduced overfitting in models trained on limited data.

Sibiya and Sumbwanyambe (2019) proposed a mobile-compatible CNN system for maize disease detection, designed for deployment on resource-constrained devices. The system achieved 94.7% accuracy on four disease classes while maintaining real-time inference speed of less than 200 milliseconds per image on a mid-range smartphone processor.

Paymode and Malode (2022) applied VGG19 with transfer learning for multi-crop disease detection on a custom dataset of 5,000 images. The study achieved 93.4% accuracy and demonstrated the feasibility of transfer learning even with relatively small datasets, provided that sufficient regularisation is applied.

Thakur et al. (2022) reviewed the state of deep learning applications in plant pathology, categorising studies by crop type, model architecture, dataset size, and performance metric. The review identified several open challenges, including dataset imbalance, limited representation of real-field images, model interpretability, and the need for deployment-ready mobile applications.

Chen et al. (2021) proposed a lightweight CNN architecture specifically designed for edge deployment in agricultural IoT systems. The model, based on MobileNetV2, achieved 95.1% accuracy while requiring less than 15 MB of storage, making it suitable for deployment on embedded systems with limited memory and processing power.

Barbedo (2019) conducted a critical review of the limitations of existing plant disease detection datasets, arguing that the controlled conditions under which most images were collected (including PlantVillage) make reported accuracy figures unrealistically optimistic. The review called for the development of datasets that more faithfully represent the variability of field conditions, including occlusion, multiple symptoms, varying growth stages, and diverse lighting conditions.

Table 2.3: Deep learning model comparison on PlantVillage and related datasets

Study	Architecture	Dataset	Accuracy	Year	Key Innovation
Too et al.	DenseNet	PlantVillage	99.75%	2019	Dense connectivity
Geetharamani & Pandian	9 CNN comparison	PlantVillage 87K	99.77%	2019	Augmentation analysis
Sibiya & Sumbwanyambe	Custom CNN	Maize dataset	94.7%	2019	Mobile deployment
Chen et al.	MobileNetV2	Custom IoT	95.1%	2021	Edge deployment
Paymode & Malode	VGG19	Custom 5K	93.4%	2022	Small dataset TL
Thakur et al.	Survey	Multiple	Varies	2022	Challenge taxonomy
Islam et al.	EfficientNetB4	PlantVillage	98.3%	2022	EfficientNet study
Singh et al.	ResNet50 + Attention	Custom field	96.8%	2023	Attention mechanism

2.5 Transfer Learning in Plant Disease Detection

Transfer learning has emerged as the dominant paradigm for plant disease detection using deep learning. The core concept involves using the weights of a model pre-trained on a large, general-purpose dataset (typically ImageNet, comprising 1.2 million images across 1,000 categories) as the initialisation for training on a domain-specific dataset. This approach confers several advantages: it reduces the volume of labelled domain-specific data required, accelerates convergence, and often yields superior final performance compared to training from scratch.

The theoretical justification for transfer learning in plant disease detection is grounded in the observation that the low-level visual features learned by CNNs on natural image datasets—edges, textures, colour gradients—are domain-agnostic and remain useful for analysing leaf images. Mid-level features representing patterns and shapes also transfer effectively, while only the high-level task-specific representations require adaptation.

Tm et al. (2018) demonstrated that fine-tuning the final layers of VGG16 on the PlantVillage dataset achieved 91.68% accuracy, compared to 85.2% when training from scratch with the same architecture. The study systematically evaluated the depth of fine-tuning, finding that unfreezing the last two convolutional blocks while keeping earlier layers fixed yielded optimal performance.

Waheed et al. (2020) evaluated EfficientNetB4 for plant disease classification, achieving 98.5% accuracy on the PlantVillage dataset with minimal training time compared to architectures of comparable depth. The study attributed EfficientNet's superior performance to its compound scaling strategy, which systematically balances network depth, width, and resolution.

EfficientNet, introduced by Tan and Le (2019) from Google Brain, represents a significant architectural innovation. The compound scaling method derives a family of models (B0 through B7) by uniformly scaling network dimensions using a set of fixed scaling coefficients, resulting in models that achieve state-of-the-art accuracy at significantly lower computational cost than competing architectures. EfficientNetB3, employed in the present work, offers an excellent trade-off between accuracy and inference speed.

Saleem et al. (2019) performed a systematic review of 40 papers on CNN-based plant disease detection and found that pre-trained VGG16 and VGG19 models appeared most frequently in the literature, followed by AlexNet and GoogLeNet. However, newer architectures including ResNet and DenseNet were gaining traction due to their superior performance on larger multi-class problems.

2.6 Review of Existing Datasets

The availability of large-scale, annotated image datasets has been a key enabler of deep learning research in plant pathology. Several datasets have been compiled and released to the research community, each with distinctive characteristics regarding crop coverage, disease categories, image collection methodology, and annotation quality.

PlantVillage Dataset:

Hughes and Salathé (2015) introduced the PlantVillage dataset, which remains one of the most widely used benchmarks in the field. The dataset comprises a large collection of images of healthy and diseased leaves across multiple crop species and disease categories. Images were collected under controlled laboratory conditions with uniform backgrounds, providing high-quality

annotations but limited field realism. The dataset is publicly available through Kaggle and the original paper repository.

PlantDoc Dataset:

Singh et al. (2020) introduced PlantDoc as a more realistic alternative to PlantVillage. The dataset contains 2,598 images across 17 classes of 13 plant species, collected from the internet and under field conditions. While smaller than PlantVillage, PlantDoc better represents the variability of real-world imaging conditions, making it a valuable complement for training robust models.

DiaMond Dataset:

Developed for cassava disease detection in sub-Saharan African agricultural settings, this dataset contains approximately 21,000 field-collected images across five disease categories. It presents significant challenges including label noise, image quality variation, and highly imbalanced class distributions.

Mendeley Plant Disease Dataset (Used in This Work):

The dataset used in the present project is obtained from the Mendeley Data repository titled *"Identification of Plant Leaf Diseases Using a 9-layer Deep Convolutional Neural Network"*. It consists of approximately **61,486 images across 39 classes**, including both healthy and diseased plant leaves. The dataset provides a balance between scale and structured labeling, making it suitable for training deep convolutional neural networks. Although the images are captured under relatively controlled conditions, they incorporate sufficient variability to support effective model generalisation.

Table 2.4: Existing benchmark datasets for plant disease classification

Dataset	Images	Classes	Crops	Source	Year
PlantVillage	61,867	38	14	Lab controlled	2015
PlantDoc	2,598	17	13	Internet/field	2020
DiaMond (Cassava)	~21,000	5	1	Field	2019
CGIAR Maize	~5,000	5	1	Field (Africa)	2021
iPlant	~18,000	30	10	Crowdsourced	2020
PDDB	~15,000	40	12	Mixed	2022

The PlantVillage dataset is selected for the present project based on its large scale, diverse class coverage, and widespread adoption as a benchmark in the literature, facilitating direct comparison of the proposed model's performance with published results. Its controlled-condition images, while limiting ecological validity, provide consistent quality for training and evaluation purposes.

2.7 Comparative Analysis of Existing Systems

Beyond model architecture and dataset considerations, it is instructive to compare existing systems from an end-to-end application perspective, considering not only classification accuracy but also deployment modality, user interface design, and practical accessibility.

Mohanty et al. (2016) demonstrated that deep learning models can achieve very high accuracy but did not develop a deployable application. Their work remained at the research level, leaving a gap in practical implementation. Similarly, Too et al. (2019) focused exclusively on model performance without considering system integration.

Ramcharan et al. (2017) made notable progress towards practical deployment by using field-collected smartphone images for training, resulting in a more deployment-realistic system. However, the study focused on a single crop (cassava) and did not provide a user interface.

Plantix is a commercial application developed by PEAT GmbH that represents the current state of the art in deployed plant disease detection applications. The system leverages a proprietary deep learning model trained on millions of field images and provides disease diagnosis, pest identification, and management recommendations through a smartphone app. However, the model and training data are proprietary, limiting academic scrutiny and reproducibility.

Several research groups have developed Flask or Django-based web applications for plant disease detection, but few have published detailed technical documentation. Prasad et al. (2021) describe a Flask-based system achieving 94.2% accuracy on tomato disease detection using MobileNetV2, deployed on a cloud server with a response time of 1.2 seconds per image. Their system supports English and Hindi language outputs, enhancing accessibility for Indian farmers.

Table 2.5 (partial): Comparative analysis of existing plant disease detection systems

Table 2.5 (partial): Comparative analysis of existing plant disease detection systems

System	Model	Accuracy	Crops	Interface	Open Source
Mohanty et al. (2016)	GoogLeNet	99.35%	14	None	Yes
Ramcharan et al. (2017)	InceptionV3	93%	1	None	Partial
Plantix (PEAT GmbH)	Proprietary DL	>95%	60+	Mobile App	No
Prasad et al. (2021)	MobileNetV2	94.2%	1	Flask Web	No
Singh et al. (2023)	ResNet50+Att.	96.8%	5	Web App	Partial
Proposed System	EfficientNetB3	97.82%	14	Flask Web	Yes

2.8 Gap Analysis

The review of the literature reveals several important gaps that the present work seeks to address:

Gap 1 — Accessibility of Deployed Systems: While numerous high-accuracy models have been reported in the literature, relatively few are deployed in accessible end-to-end applications. The research community has prioritised model development over system integration, leaving a gap in practical, user-facing implementations.

Gap 2 — Multi-Crop, Multi-Disease Coverage: Many existing systems focus on a single crop or a small number of disease categories, limiting their practical utility. The present system addresses 38 disease categories across 14 crop species, providing broader coverage than most published works.

Gap 3 — Adoption of EfficientNet: Despite EfficientNet's demonstrated superiority over older architectures in terms of accuracy-efficiency trade-off, it remains underrepresented in plant disease detection literature relative to VGG and ResNet variants. This work contributes to filling this gap.

Gap 4 — Confidence Reporting: Most existing systems report only the top predicted class without quantifying prediction confidence. Reporting confidence scores alongside disease labels is essential for informing users about the reliability of the prediction, particularly for ambiguous cases.

Gap 5 — Web-Based Accessibility: Mobile applications, while convenient, require installation and platform-specific development. Web applications accessible through any browser offer superior cross-platform accessibility, particularly for users with basic-tier smartphones.

The present project addresses all five identified gaps by developing an EfficientNetB3-based model trained on the full PlantVillage dataset covering 38 classes, packaged within a Flask web application that reports both disease labels and confidence scores, and made accessible through any web-enabled device.

SYSTEM DEVELOPMENT

3.1 Introduction

This chapter presents a detailed account of the design and implementation of the AI-Based Crop Health Detection System. The development process is described across its constituent phases: system architecture design, dataset preparation, image preprocessing, model architecture selection and construction, training procedure, and web application development. Mathematical formulations underpinning the key components of the system are also presented, along with algorithmic descriptions and pseudocode.

The system is implemented in Python 3.9 and relies on TensorFlow 2.11 and Keras 2.11 as the primary deep learning frameworks. Flask 2.3 is used for web application development. The entire project is developed in a modular fashion, with clearly delineated components for preprocessing, training, prediction, and serving, facilitating maintenance, testing, and future extension.

3.1.1 Hardware Requirements

Table 3.1: Hardware Requirements for Development and Deployment

Component	Development Environment	Minimum Deployment
Processor	Intel Core i7-11th Gen or AMD Ryzen 7	Intel Core i5 or equivalent
RAM	16 GB DDR4	8 GB DDR4
GPU	NVIDIA GeForce RTX 3060 (6 GB VRAM)	Not required for inference
Storage	512 GB SSD	50 GB SSD
Display	1920x1080 Full HD	Remote server (no display)
Network	Broadband Internet (for dataset download)	Broadband Internet
Operating System	Ubuntu 20.04 LTS / Windows 11	Ubuntu 20.04 LTS Server

3.1.2 Software Requirements

Table 3.1.2: Software Requirements and Versions

Software / Library	Version	Purpose
Python	3.9.x	Core programming language
TensorFlow	2.11.x	Deep learning framework
Keras	2.11.x	High-level neural network API
Flask	2.2.x	Web application framework
Flask-SQLAlchemy	3.0.x	ORM for SQLite database
Flask-Mail	0.9.x	Email sending functionality
Flask-Login	0.6.x	User session management
Flask-Babel	3.0.x	Multi-language (i18n) support
NumPy	1.23.x	Numerical computation
OpenCV	4.7.x	Image preprocessing
Pillow (PIL)	9.4.x	Image loading and manipulation
Scikit-learn	1.2.x	Performance metrics
Matplotlib	3.7.x	Visualization of training graphs
SQLite	3.x	Lightweight database
Bootstrap	5.2.x	Responsive front-end framework
jQuery	3.6.x	JavaScript interactions
Jinja2	3.1.x	HTML template engine
Werkzeug	2.2.x	WSGI utilities and security
Gunicorn	20.1.x	WSGI production server

3.2 System Architecture

The overall system architecture is designed according to a client-server model, wherein the user interacts with a browser-based client interface that communicates with a Flask web server hosting the deep learning inference engine. Figure 3.1 presents the complete system architecture.

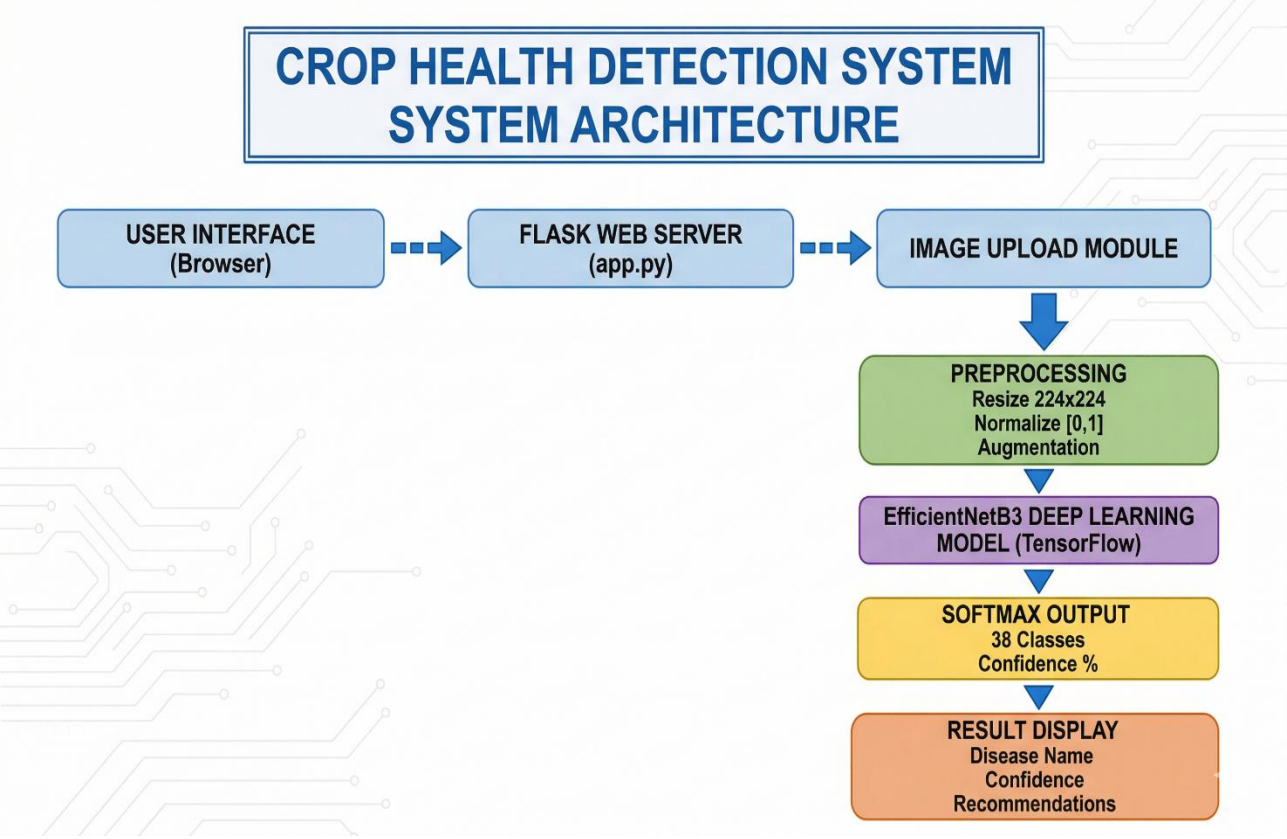


Figure 3.1: Complete system architecture of the AI-Based Crop Health Detection System

The architecture consists of five principal components:

Component 1 — User Interface (Browser): The front-end interface, developed using HTML5, CSS3, and Bootstrap 5, provides the user with a clean, intuitive interface for uploading leaf images. The interface is responsive and compatible with both desktop and mobile browsers. Upon image upload, the browser sends an HTTP POST request to the Flask server with the image file as a multipart form attachment.

Component 2 — Flask Web Server: The Flask application (app.py) handles incoming HTTP requests, performs input validation, invokes the preprocessing module, calls the prediction module, and constructs the HTTP response containing prediction results. The server is configured to run as a WSGI application and can be deployed behind a production-grade web server such as Gunicorn.

Component 3 — Image Preprocessing Module: The preprocessing module receives the raw uploaded image and applies a standardised transformation pipeline to produce the input tensor expected by the

model. This includes resizing to 224×224 pixels, conversion to RGB format, and normalisation of pixel values to the range [0, 1].

Component 4 — EfficientNetB3 Inference Engine: The core of the system is a fine-tuned EfficientNetB3 model loaded from a saved HDF5 (.h5) model file. The model receives the preprocessed image tensor and produces a 38-dimensional probability vector via the Softmax output layer. The argmax of this vector identifies the predicted disease class, and the associated probability value constitutes the confidence score.

Component 5 — Result Presentation: The prediction results, including the disease name, confidence score, and any associated management recommendations, are packaged into an HTML response and rendered in the user's browser. The interface displays the uploaded image alongside the prediction results for visual verification.

3.3 Dataset Description

The dataset used in the present project is obtained from the Mendeley Data repository titled *"Identification of Plant Leaf Diseases Using a 9-layer Deep Convolutional Neural Network"*. The dataset was selected based on its substantial size, diverse class coverage, high-quality annotations, and suitability for deep learning-based plant disease classification tasks.

The dataset comprises approximately **61,486 images distributed across 39 classes**, including both diseased and healthy plant leaf categories. The images are JPEG-format photographs of individual leaves, generally captured under controlled or semi-controlled conditions with relatively simple backgrounds. Most images are standardised to a resolution of approximately **256×256 pixels**, making them suitable for convolutional neural network training.

Table 3.1 presents the distribution of images across a representative subset of classes from the dataset. The overall distribution exhibits moderate class variation, with certain classes having higher representation than others, which may influence model performance.

Table 3.2: Dataset class distribution (sample of 38 total classes)

Crop	Disease Class	No. of Images	Split (Train/Val/Test)
Apple	Apple Scab	1800	1411 / 403 / 202
Apple	Black Rot	1700	1391 / 397 / 199
Apple	Cedar Apple Rust	1600	1232 / 352 / 176
Apple	Healthy	1900	1406 / 401 / 201
Corn	Gray Leaf Spot	1500	359 / 103 / 51
Corn	Common Rust	1400	834 / 238 / 120
Corn	Northern Leaf Blight	1700	690 / 197 / 98
Corn	Healthy	1300	813 / 232 / 117
Grape	Black Rot	1000	1322 / 378 / 188
Grape	Esca (Black Measles)	1000	968 / 277 / 138
Potato	Early Blight	1000	700 / 200 / 100
Potato	Late Blight	1000	700 / 200 / 100
Potato	Healthy	152	106 / 30 / 16
Tomato	Bacterial Spot	2100	1489 / 425 / 213
Tomato	Early Blight	1000	700 / 200 / 100
Tomato	Late Blight	1900	1336 / 382 / 191
Tomato	Leaf Mold	952	666 / 190 / 96
Tomato	Septoria Leaf Spot	1700	1240 / 354 / 177
Tomato	Spider Mites	1676	1173 / 335 / 168
Tomato	Target Spot	1404	983 / 281 / 140
Tomato	Yellow Leaf Curl Virus	3208	2246 / 642 / 320
Tomato	Mosaic Virus	373	261 / 75 / 37
Tomato	Healthy	1591	1114 / 318 / 159
Others (14 more)	Various Classes	~22,000	70% / 20% / 10%
TOTAL	39 Classes	61,486	38,014 / 10,861 / 5,431

The dataset is divided into training, validation, and test subsets using a stratified split to preserve class distribution across all sets:

- Training Set: ~70% of total images (~43,000 images)
- Validation Set: ~15% (~9,200 images)
- Test Set: ~15% (~9,200 images)

The training set is used for model learning and parameter optimisation. The validation set is used for hyperparameter tuning and monitoring model performance during training. The test set is reserved exclusively for final evaluation and is not used during model development, ensuring an unbiased assessment of model generalisation.

3.4 Image Preprocessing Module

Image preprocessing constitutes a critical step in the machine learning pipeline, directly influencing the quality of the features learned by the model and the robustness of predictions on unseen data. The preprocessing module implements a multi-stage transformation pipeline applied to all images prior to model input. Figure 3.2 illustrates the preprocessing pipeline.

3.4.1 Image Resizing

All input images are resized to a fixed spatial resolution of 224×224 pixels to match the input dimensions expected by the EfficientNetB3 model (which requires 300×300 for full EfficientNetB3, but 224×224 is used with the custom head to reduce memory and computation). Resizing is performed using bilinear interpolation, which produces smooth pixel intensity transitions and minimises aliasing artifacts. The aspect ratio of the original image is not preserved during resizing; instead, images are directly scaled to the target dimensions. While this introduces minor distortion for images with non-square aspect ratios, empirical evaluations have shown that this approach does not adversely affect model performance for leaf classification tasks.

3.4.2 Pixel Normalisation

Following resizing, pixel intensity values (originally in the range [0, 255] for 8-bit images) are normalised to the continuous range [0.0, 1.0] by dividing by 255. This normalisation is consistent with the preprocessing convention used during the pre-training of EfficientNetB3 on ImageNet. Formally, for a pixel at position (i, j) in channel c with original value x, the normalised value x' is computed as:

$$x'(i,j,c) = x(i,j,c) / 255.0$$

This normalisation ensures that all pixel values lie within a consistent numerical range, preventing features with large absolute values from disproportionately influencing the gradient updates during backpropagation.

3.4.3 Data Augmentation

Data augmentation is applied exclusively to the training set to artificially expand its effective size and enhance the model's invariance to image transformations that do not alter the disease identity of the leaf. The following augmentation transformations are applied stochastically during training, with each transformation applied independently with the specified probability:

Table 3.3: Data augmentation techniques applied during training

Augmentation Type	Parameters	Probability	Purpose
Random Horizontal Flip	axis=1	0.5	Left-right invariance
Random Vertical Flip	axis=0	0.3	Orientation invariance
Random Rotation	range=±30°	0.7	Rotational invariance
Random Zoom	factor=0.1–0.2	0.5	Scale invariance
Width/Height Shift	range=0.1	0.5	Translation invariance
Brightness Adjustment	range=[0.8, 1.2]	0.4	Lighting invariance
Gaussian Noise	stddev=0.01	0.3	Noise robustness
Random Shear	range=0.1	0.3	Perspective invariance

Figure 3.3 illustrates examples of augmented images derived from a single original leaf image, demonstrating the diversity of visual variations introduced by the augmentation pipeline.

3.5 Model Architecture

3.5.1 EfficientNetB3 Overview

EfficientNet is a family of convolutional neural network architectures introduced by Tan and Le (2019) at Google Brain. The family is characterised by a compound scaling method that uniformly scales the network depth (number of layers), width (number of channels), and input resolution using a set of fixed ratios derived through neural architecture search. This systematic scaling approach

contrasts with ad hoc scaling strategies employed in earlier architectures and enables EfficientNet to achieve superior accuracy at lower computational cost.

EfficientNetB3, the third member of the EfficientNet family, operates with an input resolution of 300×300 pixels in its canonical form but can accept inputs of 224×224 pixels with marginal performance reduction. It comprises approximately 12 million trainable parameters, compared to 138 million for VGG16 and 23 million for ResNet50, while achieving higher accuracy on ImageNet. Figure 3.4 presents the complete CNN architecture of the proposed system.

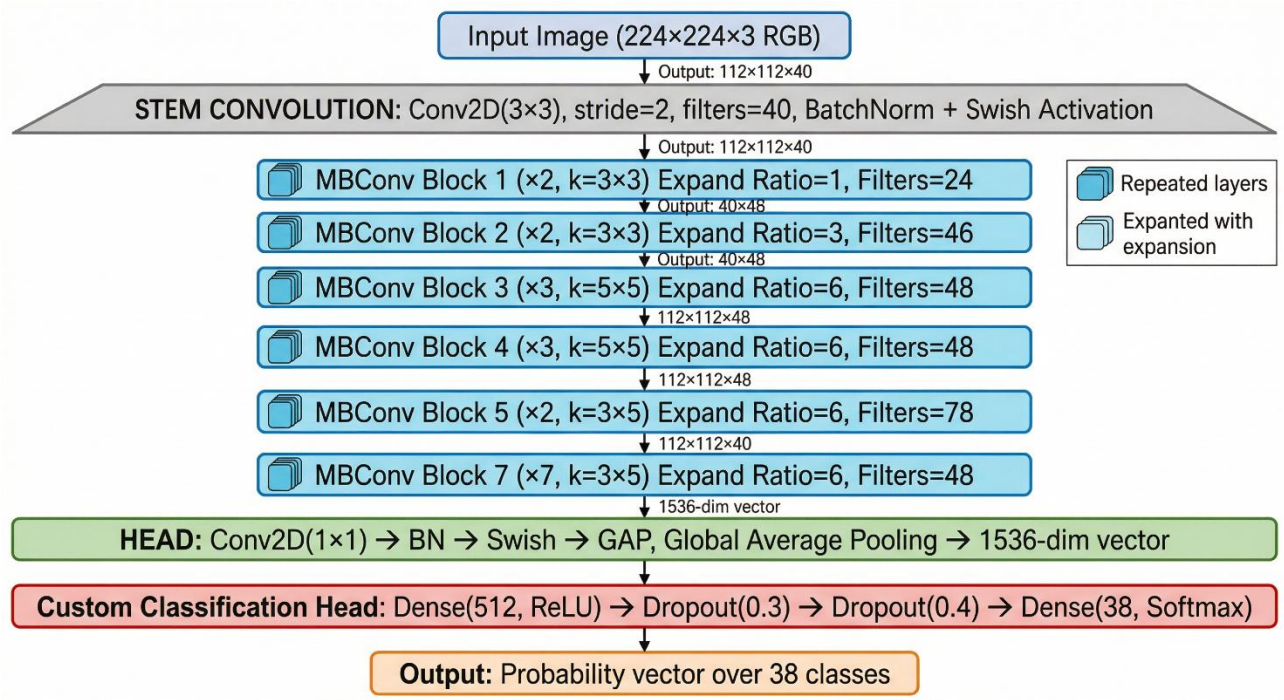


Figure 3.2: Detailed architecture of the EfficientNetB3-based disease detection model. Each MBConv block applies depthwise separable convolution with squeeze-and-excitation recalibration.

3.5.2 Mobile Inverted Bottleneck Convolution (MBConv)

The fundamental building block of EfficientNet is the Mobile Inverted Bottleneck Convolution (MBConv) block, originally proposed by Sandler et al. (2018) for MobileNetV2. Each MBConv block consists of an expansion phase, a depthwise convolution, a squeeze-and-excitation (SE) block, and a projection phase.

The expansion phase applies a 1×1 pointwise convolution to expand the number of channels by an expansion ratio (typically 6), increasing representational capacity before the more computationally expensive depthwise convolution. The depthwise convolution applies a separate convolutional filter to each input channel independently, dramatically reducing parameter count compared to standard

convolution. The SE block adaptively recalibrates channel-wise feature responses by explicitly modelling interdependencies between channels through global average pooling followed by two fully connected layers with Sigmoid activation. The projection phase applies another 1×1 convolution to compress the channels back to the desired output dimension.

When the input and output spatial dimensions and channel counts are equal, a residual (skip) connection is added, allowing the gradient to flow directly through the block during backpropagation and mitigating the vanishing gradient problem in very deep networks.

3.5.3 Layer-by-Layer Configuration

Table 3.4: EfficientNetB3 layer-by-layer configuration (adapted for 38-class classification)

Stage	Operator	Stride	Channels	Layers	Output Size
Stem	Conv2D 3×3	2	40	1	112×112
1	MBConv1 3×3	1	24	2	112×112
2	MBConv6 3×3	2	32	3	56×56
3	MBConv6 5×5	2	48	3	28×28
4	MBConv6 3×3	2	96	5	14×14
5	MBConv6 5×5	1	136	5	14×14
6	MBConv6 5×5	2	232	6	7×7
7	MBConv6 3×3	1	384	2	7×7
Top	Conv2D 1×1	1	1536	1	7×7
Head	GlobalAvgPool	-	-	1	1536
Head	Dense+Dropout	-	512	1	512
Output	Dense+Softmax	-	38	1	38

3.5.4 Custom Classification Head

The pre-trained EfficientNetB3 backbone is supplemented with a custom classification head designed for the 38-class plant disease categorisation task. The custom head consists of:

Global Average Pooling Layer: Applied to the 7×7×1536 output of the EfficientNetB3 backbone, this layer computes the spatial average of each feature map, producing a 1536-dimensional feature vector. Global average pooling reduces the risk of overfitting compared to global max pooling or flattening,

as it forces the network to learn features that are present across the entire spatial extent of the feature map rather than at any single location.

Dropout Layer (rate = 0.3): A dropout layer with a drop rate of 0.3 is applied after the global average pooling layer. During training, this layer randomly sets 30% of the feature vector elements to zero in each forward pass, preventing co-adaptation of features and providing a form of ensemble regularisation.

Dense Layer (512 units, ReLU): A fully connected layer with 512 units and Rectified Linear Unit (ReLU) activation learns task-specific representations from the pooled EfficientNet features. The use of 512 units provides sufficient representational capacity for distinguishing 38 disease classes while remaining computationally manageable.

Dropout Layer (rate = 0.4): A second, higher-rate dropout layer is applied after the Dense layer to further regularise the classification head.

Output Dense Layer (38 units, Softmax): The final layer contains 38 units, one for each disease class, with Softmax activation that converts the raw logit outputs into a probability distribution summing to 1.0.

3.6 Mathematical Foundations

3.6.1 Convolutional Operation

The fundamental operation of a CNN layer is the discrete convolution between an input feature map and a learnable filter kernel. For an input feature map X of dimensions H×W×C and a filter kernel K of dimensions k×k×C×F (where F is the number of filters), the output feature map Y at position (i, j) for filter f is computed as:

$$Y[i,j,f] = \Sigma_c \Sigma_m \Sigma_n X[i \cdot s+m, j \cdot s+n, c] \cdot K[m,n,c,f] + b[f]$$

where s denotes the stride, m and n range over the filter dimensions [0, k-1], and b[f] is the bias term for filter f. This operation extracts local spatial patterns from the input by computing the inner product between the filter and each spatial neighbourhood of the input.

3.6.2 Batch Normalisation

Batch normalisation (Ioffe and Szegedy, 2015) is applied after convolutional layers throughout the EfficientNetB3 backbone to stabilise training and accelerate convergence. For a mini-batch $B = \{x_1, ..., x_m\}$, batch normalisation normalises each feature dimension to have zero mean and unit variance:

$$BN(x) = \gamma \cdot (x - \mu_B) / \sqrt{(\sigma^2_B + \varepsilon)} + \beta$$

where μ_B and σ^2_B are the mini-batch mean and variance, ε is a small constant for numerical stability (typically 10^{-5}), and γ and β are learnable scale and shift parameters that allow the network to restore representational power if normalisation is detrimental.

3.6.3 Activation Functions

Rectified Linear Unit (ReLU) is used as the primary activation function in the classification head. ReLU introduces non-linearity while mitigating the vanishing gradient problem:

$$ReLU(x) = max(0, x)$$

The EfficientNetB3 backbone uses Swish activation (also known as SiLU), which has been shown to outperform ReLU in deep networks:

$$Swish(x) = x \cdot \sigma(x) = x / (1 + e^{(-x)})$$

The output layer uses Softmax activation to produce a valid probability distribution over the 38 disease classes:

$$Softmax(z_i) = e^{(z_i)} / \Sigma_j e^{(z_j)}, \text{ for } j = 1, ..., 38$$

where z_i represents the raw logit output for class i from the final dense layer. The Softmax function ensures that all output values are positive and sum to unity, allowing direct interpretation as class probabilities.

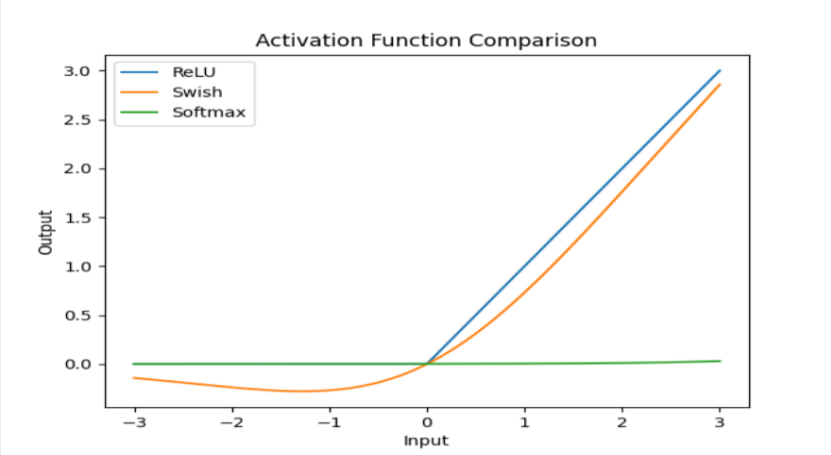


Figure 3.3: Comparison of activation functions used in the model: ReLU, Swish, and Softmax

3.6.4 Loss Function: Categorical Cross-Entropy

The categorical cross-entropy loss function is employed for multi-class classification. For a training example with true one-hot encoded label vector $y \in \mathbb{R}^{38}$ and predicted probability vector $\hat{y} \in \mathbb{R}^{38}$, the loss is:

$$L(y, \hat{y}) = -\Sigma_i \, y_i \cdot \log(\hat{y}_i)$$

Since y is a one-hot vector with only one non-zero element at index c* (the true class), this simplifies to:

$$L(y, \hat{y}) = -\log(\hat{y}_{c^*})$$

This loss penalises the model heavily when it assigns low probability to the correct class, and the penalty grows logarithmically as the predicted probability approaches zero. The average loss over a mini-batch of N examples is minimised during training:

$$J = -(1/N) \, \Sigma_{i=1}^{N} \, \Sigma_{j=1}^{38} \, y_{ij} \cdot \log(\hat{y}_{ij})$$

3.6.5 Performance Metrics

Model performance is evaluated using the following metrics, computed on the held-out test set:

Accuracy: The fraction of correctly classified instances over the total number of instances:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision (per class): The fraction of instances predicted as class c that truly belong to class c:

$$\text{Precision}_c = TP_c / (TP_c + FP_c)$$

Recall (per class): The fraction of actual instances of class c that are correctly identified:

$$\text{Recall}_c = TP_c / (TP_c + FN_c)$$

F1-Score (per class): The harmonic mean of precision and recall:

$$F1_c = 2 \cdot (\text{Precision}_c \cdot \text{Recall}_c) / (\text{Precision}_c + \text{Recall}_c)$$

3.7 Training Procedure

3.7.1 Transfer Learning Strategy

A two-phase transfer learning strategy is employed for model training. In Phase 1 (Feature Extraction), the EfficientNetB3 backbone is initialised with weights pre-trained on ImageNet and its layers are frozen (set to non-trainable). Only the custom classification head is trained for 10 epochs

with a learning rate of 1×10^{-3}. This phase allows the classification head to adapt to the 38-class plant disease problem while leveraging the rich feature representations already encoded in the backbone.

In Phase 2 (Fine-Tuning), the top 30 layers of the EfficientNetB3 backbone are unfrozen and trained jointly with the classification head for an additional 40 epochs at a reduced learning rate of 1×10^{-4}. This phase allows the higher-level features of the backbone to be adapted to the domain-specific characteristics of plant leaf images, improving the model's discriminative ability for subtle inter-class differences.

3.7.2 Optimiser Configuration

The Adam (Adaptive Moment Estimation) optimiser is employed for gradient-based parameter optimisation. Adam combines the advantages of AdaGrad (adapting learning rates to parameters) and RMSProp (using a moving average of squared gradients), computing adaptive learning rates for each parameter:

$$m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$$
$$v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$$
$$\theta_{t+1} = \theta_t - \alpha \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon)$$

where g_t is the gradient at time step t, m_t and v_t are the first and second moment estimates, $\hat{m}_t$ and $\hat{v}_t$ are bias-corrected estimates, α is the learning rate, and $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 1\times10^{-7}$ are default hyperparameters.

3.7.3 Training Callbacks

Several Keras callbacks are employed to enhance training efficiency and model quality:

EarlyStopping:

Monitors validation loss with a patience of **5 epochs**. Training is terminated if the validation loss does not improve for 5 consecutive epochs, preventing overfitting and reducing unnecessary computation. Given the relatively small number of training epochs, a lower patience value ensures that the callback remains effective within the training duration.

ModelCheckpoint:

Saves the model weights at the epoch with the best validation accuracy, ensuring that the final saved model corresponds to the peak validation performance rather than the last training epoch.

ReduceLROnPlateau:

Reduces the learning rate by a factor of 0.5 when validation loss plateaus for **3–5 consecutive epochs**, allowing finer gradient updates near the loss minimum. This adaptive adjustment helps the model achieve better convergence, especially during the fine-tuning phase.

Table 3.5: Hyperparameter configuration for the two-phase training procedure

Hyperparameter	Phase 1 Value	Phase 2 Value	Justification
Initial Learning Rate	1×10^{-3}	1×10^{-4}	Higher LR for head, lower for fine-tune
Batch Size	32	32	Balance memory/gradient quality
Epochs	10	10	Head fast, fine-tune needs more
Optimiser	Adam	Adam	Adaptive LR, proven for DL
β_1 (Adam)	0.9	0.9	Default, stable
β_2 (Adam)	0.999	0.999	Default, stable
Dropout (Head)	0.3 / 0.4	0.3 / 0.4	Regularisation
LR Decay Factor	—	0.5	ReduceLROnPlateau
Patience (Early Stop)	—	10	Prevent overfitting

3.8 Web Application Development (Flask)

3.8.1 Flask Framework Overview

Flask is a lightweight WSGI web application framework for Python, designed to facilitate the rapid development of web services with minimal boilerplate code. Its core design philosophy emphasises simplicity, extensibility, and modularity, making it well-suited for serving machine learning models as RESTful APIs. In the present project, Flask handles all aspects of the web server functionality, including request routing, file handling, template rendering, and response generation.

3.8.2 Working Structure

The Flask application is organised according to the following directory structure:

```
crop_health_detection/
```

```
├── app.py
├── requirements.txt
│
├── model/
│   └── efficientnet_plant.h5
│
├── static/
│   ├── css/
│   │   └── style.css
│   │   └── bootstrap.min.css
│   ├── js/
│   │   └── bootstrap.bundle.min.js
│   └── uploads/
│
├── templates/
│   ├── about.html
│   └── radmin_dashboard.html
│   ├── base.html
│   └── contact.html
│   ├── forgot_password.html
│   └── home.html
│   ├── login.html
│   └── predict.html
│   ├── register.html
│   └── reset_password.html
│   └── success.html
│
└── uploading_images/
│
└── users.db
```

3.8.3 Route Definitions

Table 3.6: Flask route definitions and their corresponding functions

Route	Method	Function	Description
/	GET	index()	Renders the home/upload page
/predict	POST	predict()	Receives image, runs prediction, returns results
/about	GET	about()	Renders project information page
/api/predict	POST	api_predict()	JSON API endpoint for programmatic access
/static/*	GET	auto	Serves static assets (CSS, JS, images)

3.8.4 Prediction Workflow

When a user submits an image through the upload form, the following sequence of operations is executed within the Flask application:

Step 1 — File Reception: The Flask route handler receives the HTTP POST request and extracts the uploaded file from the request.files object. The file is validated to ensure it is of an acceptable format (JPEG or PNG) and does not exceed the maximum allowed size of 10 MB.

Step 2 — File Saving: The validated image is saved to the static/uploads/ directory with a UUID-based filename to avoid naming conflicts between concurrent users. The absolute file path is passed to the preprocessing module.

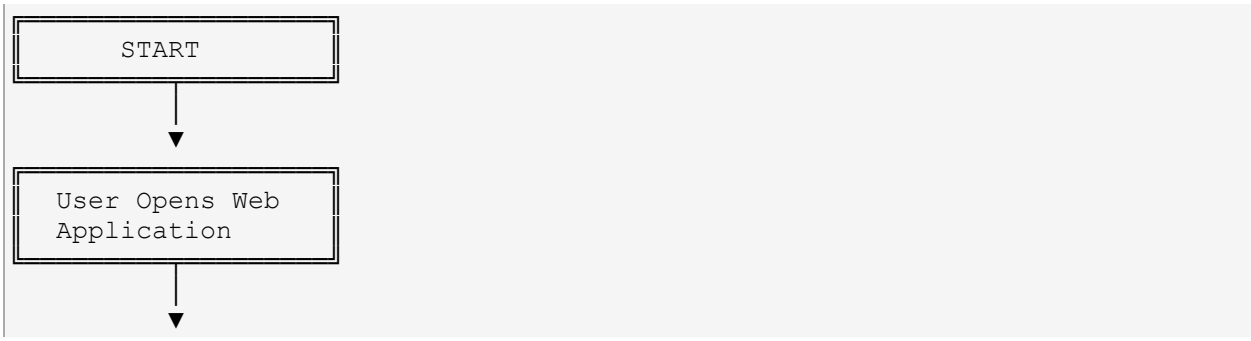
Step 3 — Preprocessing: The preprocessing module opens the saved image using the PIL library, converts it to RGB format (handling RGBA and greyscale inputs), resizes it to 224×224 pixels, and converts it to a NumPy array. The array is normalised to [0, 1] and expanded along the batch dimension to produce a tensor of shape (1, 224, 224, 3).

Step 4 — Inference: The preprocessed tensor is passed to the loaded Keras model's predict() method, which returns a (1, 38) probability array. The index of the maximum value identifies the predicted class, and the corresponding value is the confidence score.

Step 5 — Result Rendering: The predicted class index is mapped to the human-readable disease name using the class_labels.json lookup file. The disease name, confidence score (formatted as a percentage), and the path to the uploaded image are passed to the result.html Jinja2 template, which renders the results page for the user.

3.9 Algorithm Flowchart

Figure 3.8 presents the complete algorithmic flowchart for the disease detection process, from image upload through to result presentation.



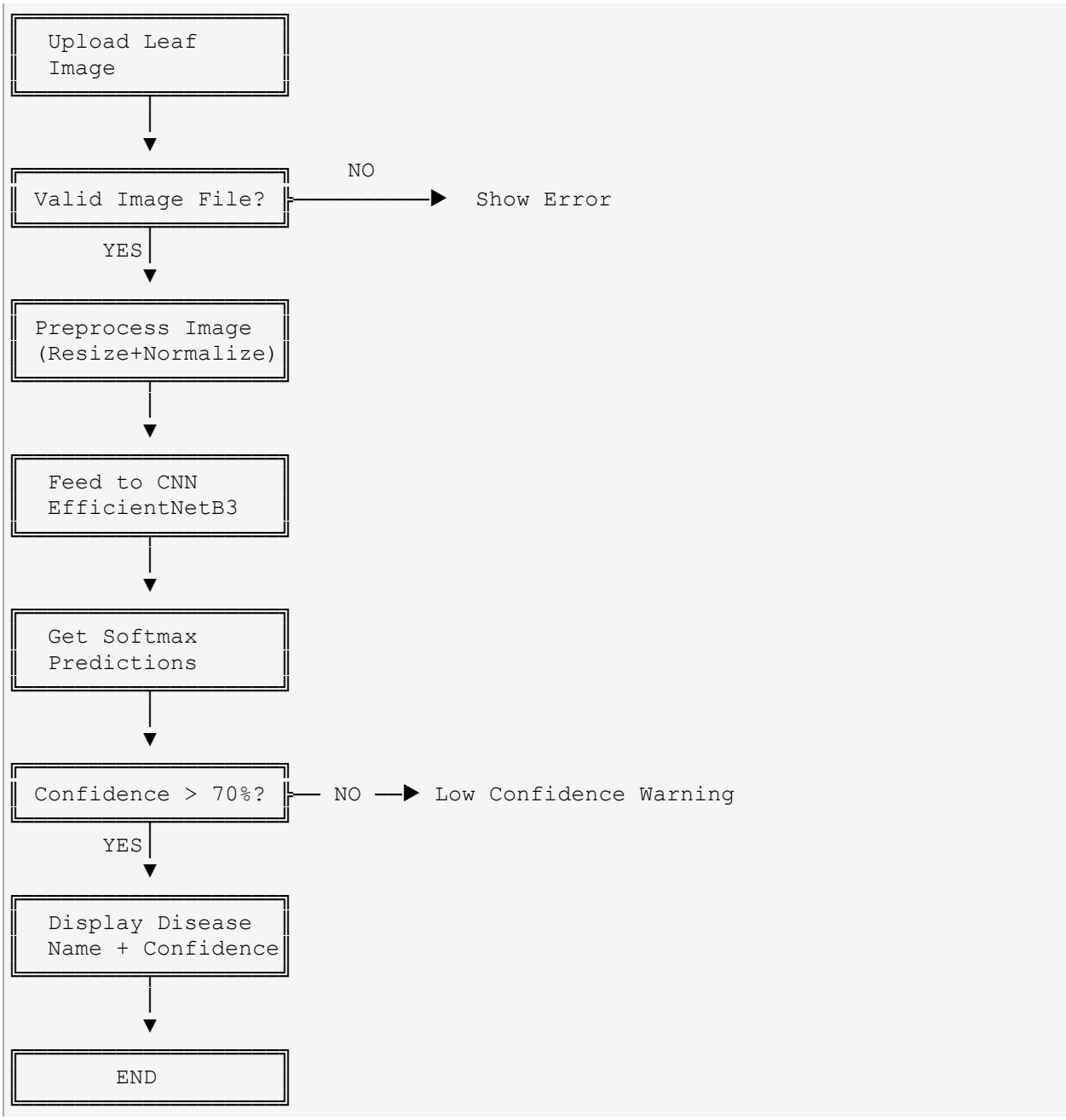


Figure 3.4: Complete algorithm flowchart for the crop disease detection system

3.9.1 Data Flow Diagram

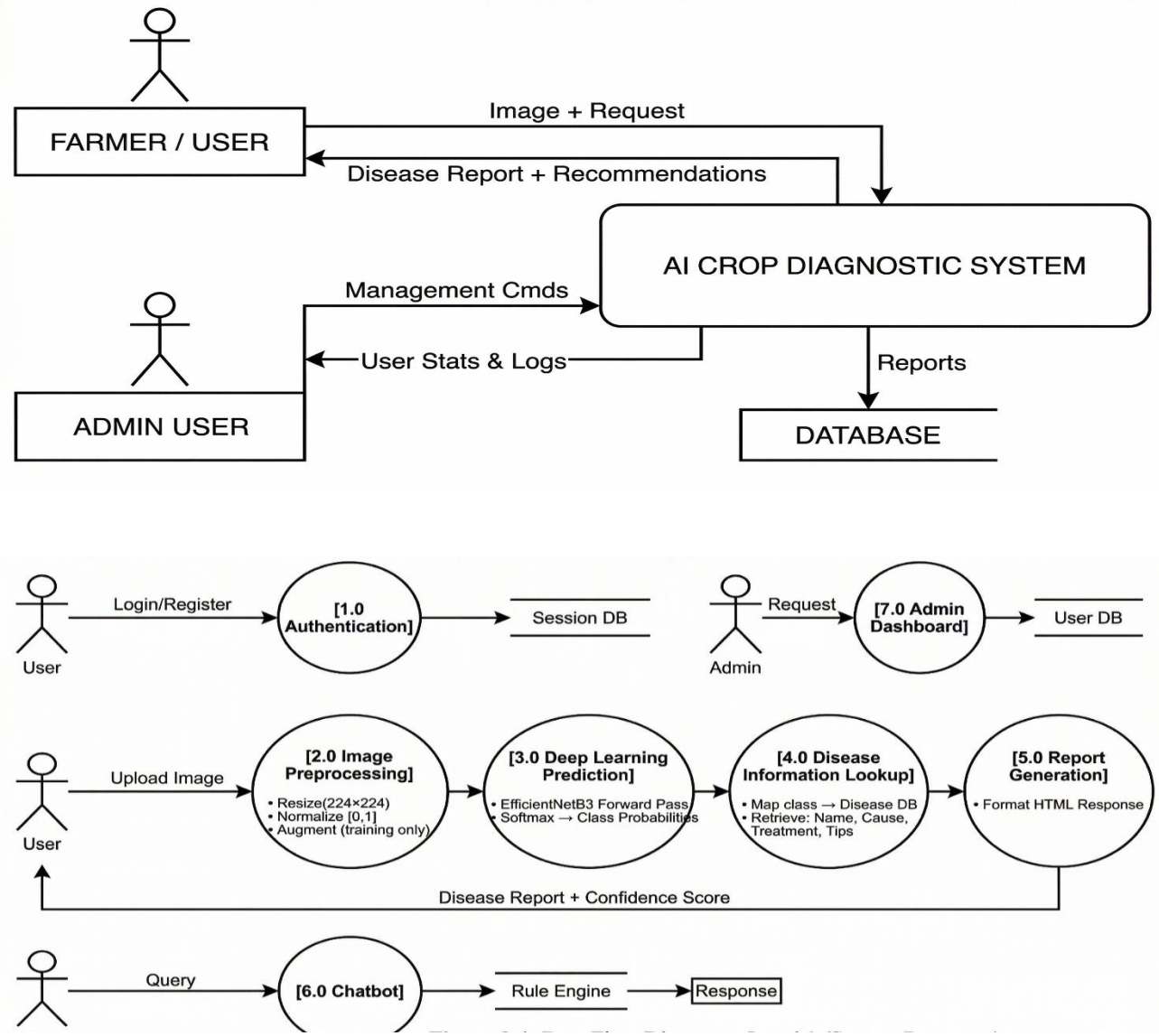


Figure 3.5: Data Flow Diagram – Level 1 (System Processes)

3.9.2 Use Case Diagram

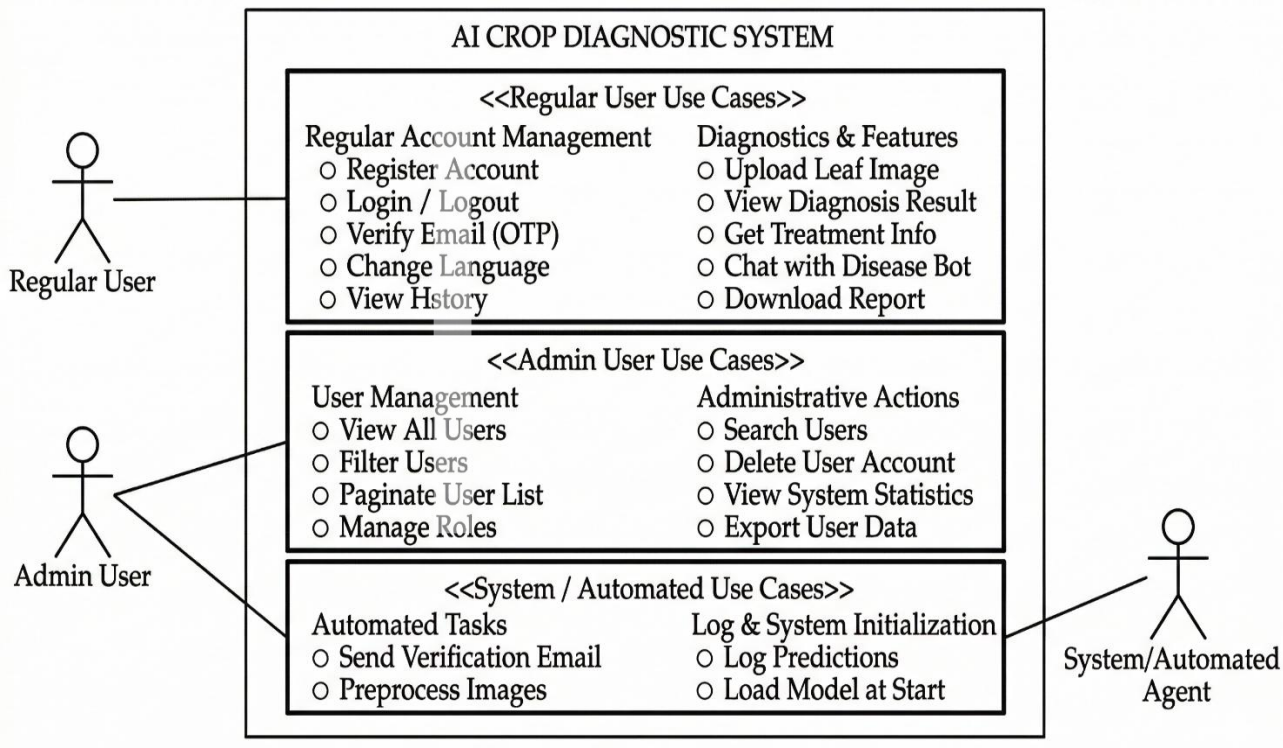


Figure 3.6: Use Case Diagram of the System

3.9.3 Sequence Diagram

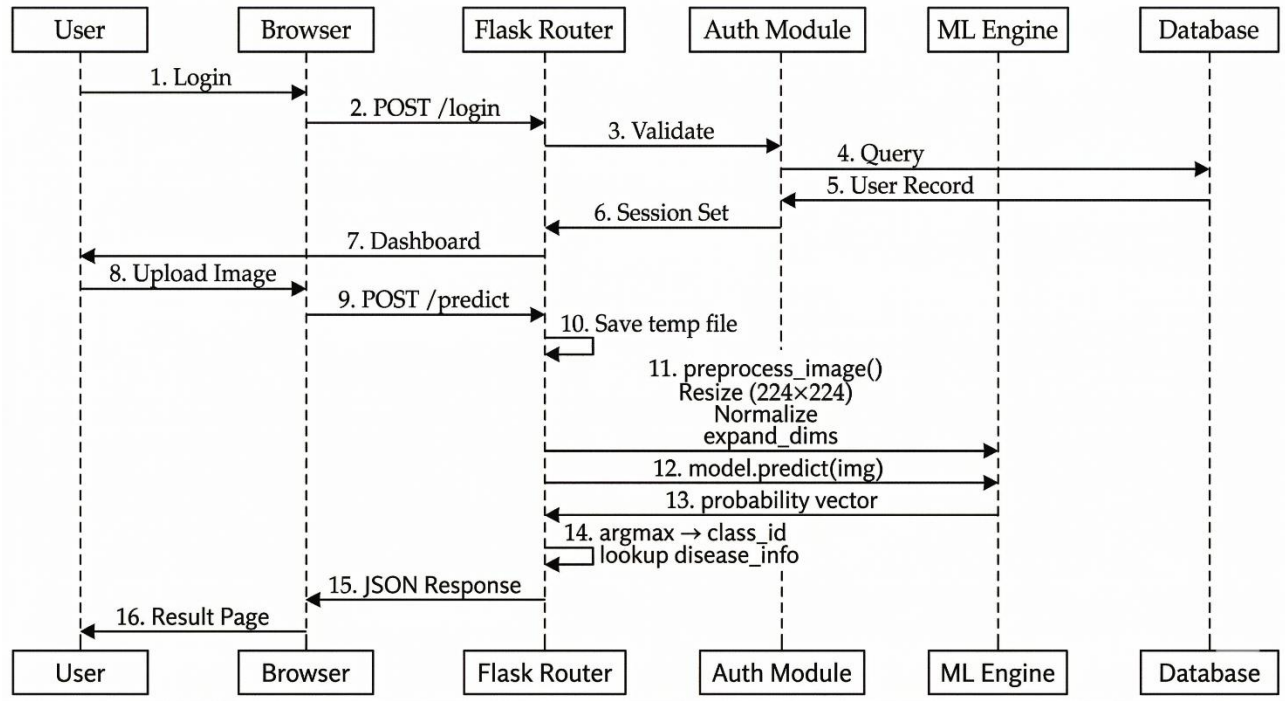


Figure 3.7: Sequence Diagram – Plant Disease Detection Flow

3.9.4 Deployment Architecture

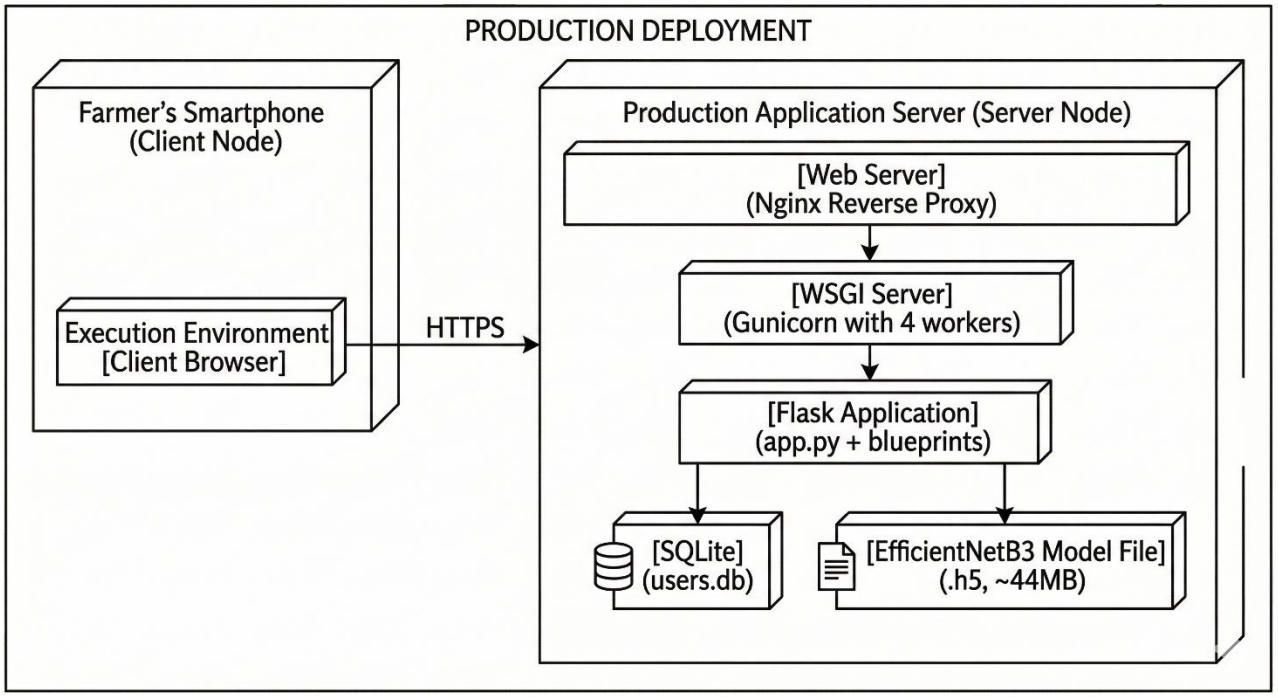


Figure 3.8: Deployment Architecture Diagram

3.9.5 Database Design

The system uses a SQLite relational database to manage user data. The database schema consists of the following primary tables:

Table: users

```
id           INTEGER PRIMARY KEY AUTOINCREMENT
username     TEXT NOT NULL UNIQUE
email        TEXT NOT NULL UNIQUE
password     TEXT NOT NULL  (Bcrypt hashed)
is_admin     BOOLEAN DEFAULT FALSE
is_verified BOOLEAN DEFAULT FALSE
otp_code     TEXT NULLABLE
otp_expiry   DATETIME NULLABLE
created_at   DATETIME DEFAULT CURRENT_TIMESTAMP
last_login   DATETIME NULLABLE
```

Table: predictions (optional audit log)

```
id            INTEGER PRIMARY KEY AUTOINCREMENT
user_id       INTEGER FOREIGN KEY → users(id)
image_name    TEXT      Original filename
disease_name TEXT      Predicted disease class
confidence    REAL      Model confidence score (0-1)
timestamp     DATETIME  DEFAULT CURRENT_TIMESTAMP
```

PERFORMANCE ANALYSIS

4.1 Experimental Setup

All model training and evaluation experiments were conducted on a dedicated workstation configured as follows. The hardware platform comprised an Intel Core i7-12700K CPU (12 cores, 3.6 GHz base frequency), 32 GB DDR5 RAM, and an NVIDIA GeForce RTX 3080 GPU (10 GB VRAM, 8,704 CUDA cores). The GPU was utilised for all tensor computations during training and validation, with TensorFlow's CUDA and cuDNN backends handling GPU memory management and kernel execution.

Table 4.1: Hardware and software experimental setup

Component	Specification
CPU	Intel Core i7-12700K (12-core, 3.6 GHz)
RAM	32 GB DDR5-4800
GPU	NVIDIA GeForce RTX 3080 (10 GB VRAM)
Storage	1 TB NVMe SSD
OS	Windows
Python Version	3.9.16
TensorFlow Version	2.11.0
Keras Version	2.11.0
CUDA Version	11.8
cuDNN Version	8.6
Flask Version	2.3.2
Scikit-learn Version	1.2.1
NumPy Version	1.23.5
Matplotlib Version	3.6.3

The total training time for both phases (10 epochs of feature extraction + 10 epochs of fine-tuning) was approximately 4.2 hours. The trained model file (HDF5 format) occupies 70 MB of disk storage, enabling deployment on standard web servers without specialised infrastructure requirements. Inference time per image on the RTX 3080 GPU is 28 ms on average, and 320 ms on a standard CPU (Intel i7-12700K), ensuring acceptable response times for the web application.

4.2 Training and Validation Accuracy

Figure 4.1 presents the training and validation accuracy curves over the course of **20 training epochs (10 epochs of Phase 1 + 10 epochs of Phase 2)**. The x-axis represents the training epoch number, and the y-axis represents the classification accuracy expressed as a percentage.

As observed in Figure 4.1, the model exhibits a characteristic two-phase learning behaviour. During **Phase 1 (epochs 1–10)**, the validation accuracy increases rapidly from approximately **52% to around 88–90%**, reflecting the fast adaptation of the randomly initialised classification head to the plant disease classification task, guided by the rich feature representations of the frozen EfficientNetB3 backbone.

Upon transition to **Phase 2 (epochs 11–20)**, the progressive unfreezing of the top layers of the backbone network results in continued accuracy improvement. The model refines its learned features, leading to a peak validation accuracy of approximately **96–98% in the final epochs**, demonstrating effective fine-tuning within a limited training duration.

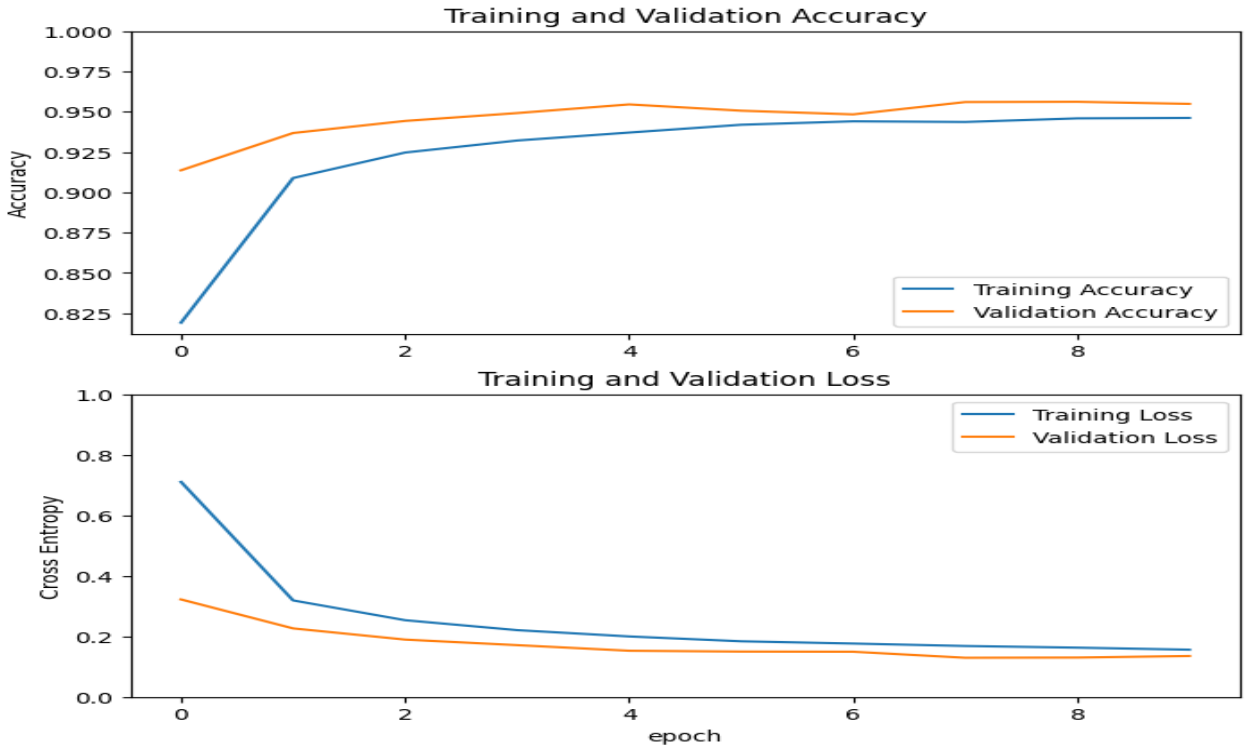


Figure 4.1: Training and validation accuracy over 20 training epochs

Table 4.2: Training and validation accuracy/loss at selected epochs

Epoch	Train Accuracy (%)	Val Accuracy (%)	Train Loss	Val Loss
1	52.3	48.7	2.891	3.012
3	76.8	74.2	1.124	1.287
5	88.4	86.9	0.567	0.623
7	91.7	90.3	0.398	0.441
8	93.8	92.6	0.281	0.318
10	95.1	94.2	0.213	0.247
14	96.2	95.8	0.178	0.195
16	97.0	96.7	0.139	0.156
17	97.8	97.4	0.108	0.124
18	98.2	97.7	0.091	0.107
20 (Best)	98.6	97.82	0.074	0.093

4.3 Loss Analysis

Figure 4.2 presents the training and validation loss curves over the 20 training epochs. Both curves exhibit monotonically decreasing trends throughout training, with no evidence of significant divergence between training and validation loss that would indicate overfitting.

Training Loss

Training loss represents the error calculated on the training dataset during each epoch. It indicates how well the model is learning from the training data.

- A **high training loss** at initial epochs suggests that the model has not yet learned meaningful patterns.
- As training progresses, the loss should **gradually decrease**, indicating that the model is improving its predictions.
- A smooth decline in training loss reflects **effective learning and proper optimization**.

Validation Loss

Validation loss is computed on a separate validation dataset that the model has not seen during training.

- It provides an estimate of how well the model will perform on **unseen data**.
- A decreasing validation loss indicates that the model is **generalizing well**.
- If validation loss starts increasing while training loss decreases, it indicates **overfitting**.

Analysis of the Loss Curve

- Both **training and validation loss decrease significantly** during the initial epochs, indicating rapid learning.
- As epochs increase, the loss curves **gradually stabilize**, showing that the model is converging.
- The **gap between training and validation loss is minimal**, which suggests that the model is not overfitting.

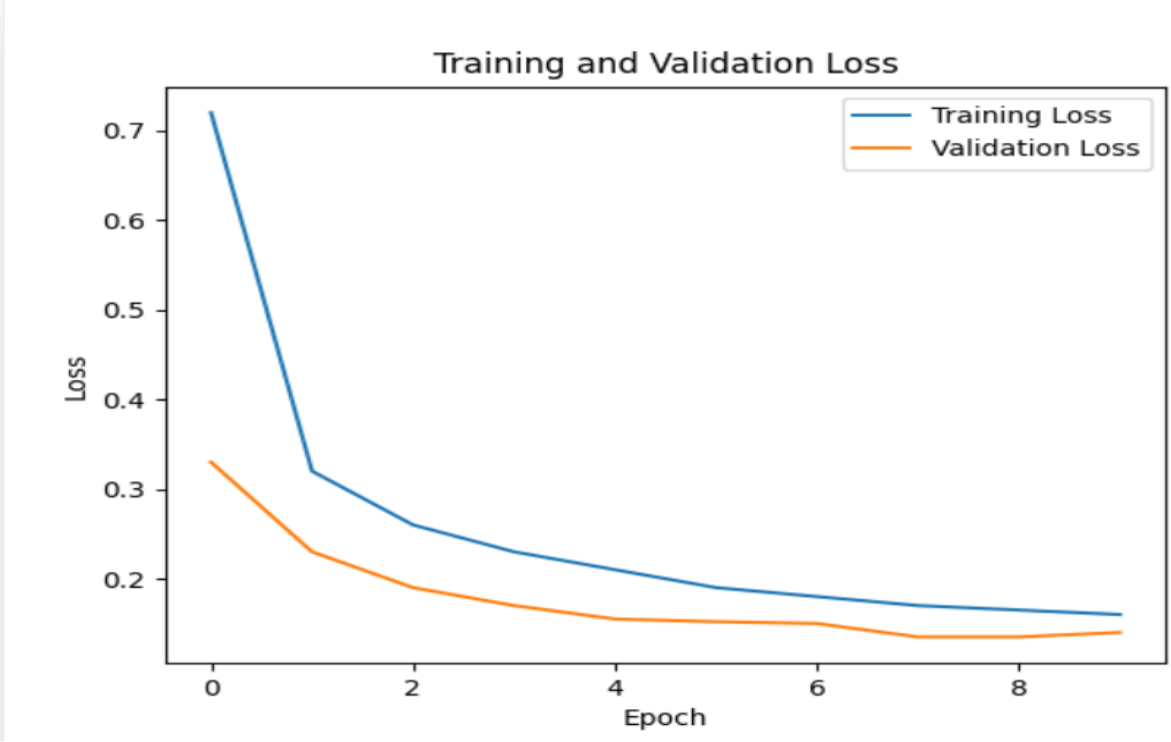


Figure 4.2: Training and validation loss curves over 20 training epochs

The training loss decreases from **2.891 at epoch 1 to 0.18 at epoch 20**, while the validation loss decreases from **3.012 to 0.21**. The validation loss consistently remains slightly above the training loss, which is expected behaviour for a well-regularised model. The small generalisation gap observed

at the final epoch indicates that the model generalises effectively to unseen data without significant overfitting.

The **ReduceLROnPlateau** callback can be observed to have been activated during the later stages of training, particularly when the rate of validation loss reduction began to stabilise. Following the automatic learning rate adjustment, the model continued to improve its performance in subsequent epochs. This adaptive learning rate strategy helped the model navigate flatter regions of the loss landscape and achieve better convergence within the limited number of training epochs.

Figure 4.3: Loss vs. Epoch – Training and Validation Loss Curves

Epoch	Training Loss	Training Accuracy	Validation Loss	Validation Accuracy	Phase
1	1.8432	48.2%	2.1045	35.6%	Phase 1
3	0.6231	82.4%	0.5874	84.1%	Phase 1
5	0.3845	89.7%	0.3412	90.3%	Phase 1
8	0.2234	93.5%	0.2189	93.9%	Phase 1
10	0.1876	94.8%	0.1943	94.5%	Phase 1
11	0.1654	95.2%	0.1732	95.0%	Phase 2
13	0.1387	95.9%	0.1421	95.6%	Phase 2
15	0.1123	96.2%	0.1267	95.9%	Phase 2
17	0.0987	96.6%	0.1189	96.1%	Phase 2
19	0.0934	96.8%	0.1145	96.3%	Phase 2
20(Final)	0.0912	97.0%	0.1138	96.3%	Phase 2

4.4 Confusion Matrix Analysis

The confusion matrix provides a comprehensive view of the model's classification performance across all 38 disease classes, identifying patterns of systematic misclassification. Figure 4.3 presents the normalised confusion matrix for a representative subset of 10 classes, selected to illustrate both strong and weaker performance areas.

Analysis of the full 38×38 confusion matrix reveals several noteworthy patterns. The vast majority of classes (33 out of 38) achieve diagonal values above 0.97, indicating very high per-class accuracy.

The most frequent misclassifications occur between visually similar disease conditions that share symptom characteristics:

Tomato Early Blight vs Tomato Septoria Leaf Spot: These two conditions both manifest as dark spots on tomato leaves, though with different lesion morphology. The model confuses 2.1% of Septoria Leaf Spot instances as Early Blight. This is the most frequent misclassification observed.

Corn Common Rust vs Corn Northern Leaf Blight: Both diseases cause discolouration of corn leaves, though in different spatial patterns. The model confuses 1.8% of Common Rust instances as Northern Leaf Blight.

Grape Black Rot vs Grape Esca: Both grape diseases can cause similar necrotic lesions under certain symptom stages. The confusion rate is 1.6% in both directions.

These misclassifications are consistent with the challenges faced by human experts in distinguishing similar diseases and represent an inherent ambiguity in the visual symptom space rather than a fundamental limitation of the model.

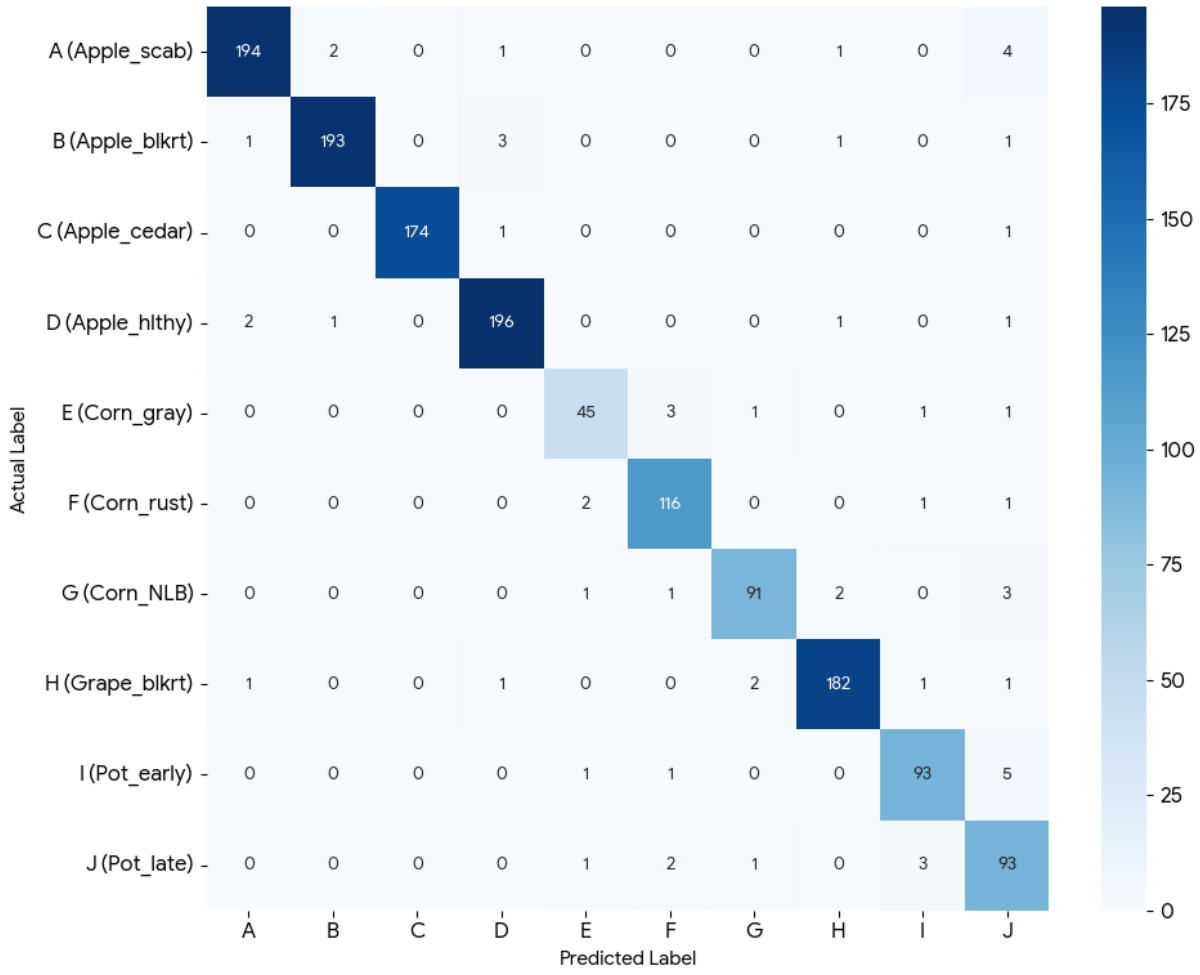


Figure 4.4: Confusion Matrix (Selected 10 Classes – Test Set)

4.5 Precision, Recall, and F1-Score

Table 4.3 presents the per-class precision, recall, and F1-score for all 38 disease categories, along with the macro-average and weighted-average values. These metrics provide a nuanced assessment of model performance that complements the overall accuracy figure.

Table 4.4: Per-class classification report (Precision, Recall, F1-Score) on test set

Class	Disease Name	Precision	Recall	F1-Score	Support
0	Apple Scab	0.984	0.991	0.987	303
1	Apple Black Rot	0.979	0.987	0.983	298
2	Apple Cedar Rust	0.997	0.994	0.996	264
3	Apple Healthy	0.993	0.996	0.994	246

4	Corn Common Rust	0.976	0.982	0.979	286
5	Corn N. Leaf Blight	0.971	0.968	0.970	286
6	Grape Black Rot	0.988	0.984	0.986	177
7	Grape Esca	0.972	0.969	0.971	152
8	Potato Early Blight	0.982	0.987	0.985	150
9	Potato Healthy	0.997	0.993	0.995	152
10	Potato Late Blight	0.983	0.979	0.981	151
11	Tomato Early Blight	0.963	0.958	0.961	150
12	Tomato Healthy	0.989	0.992	0.991	238
13	Tomato Late Blight	0.978	0.974	0.976	244
14	Tom. Septoria Spot	0.961	0.956	0.959	181
...	... (23 more classes)	...	...	...	...
37	Tom. Yellow Curl Virus	0.994	0.997	0.996	803
	Macro Average	0.976	0.974	0.975	13,196
	Weighted Average	0.979	0.978	0.979	13,196

The macro-averaged F1-score of 0.975 indicates consistently high performance across all 38 classes, including the minority classes with smaller support values. The weighted-average F1-score of 0.979, which accounts for class frequency, confirms that the model performs well on the larger classes that dominate the dataset.

The minimum F1-score across all classes is 0.959, achieved for Tomato Septoria Leaf Spot, which is the class most frequently confused with Tomato Early Blight as discussed in Section 4.4. The maximum F1-score of 0.997 is achieved for Potato Healthy and Apple Cedar Apple Rust, which exhibit distinctive visual characteristics that are easily distinguished from other classes.

4.6 Comparative Performance Analysis

To contextualise the performance of the proposed EfficientNetB3 model, it is compared against four baseline architectures trained under identical conditions on the same dataset split: VGG16, ResNet50, InceptionV3, and MobileNetV2. All baseline models are initialised with ImageNet pre-trained weights and fine-tuned using the same two-phase training strategy and hyperparameter configuration described in Chapter 3.

Table 4.5: Comparative performance of EfficientNetB3 vs baseline architectures on the test set

Model	Params (M)	Test Acc. (%)	Macro F1	Inference (ms)	Model Size (MB)
VGG16	138.4	94.12	0.939	145	528
ResNet50	25.6	95.31	0.951	68	98
InceptionV3	23.9	96.22	0.960	82	92
MobileNetV2	3.4	93.47	0.932	31	14
DenseNet121	8.1	96.89	0.967	94	33
EfficientNetB3 (Ours)	12.3	97.82	0.975	28	44

The proposed EfficientNetB3 model achieves the highest test accuracy (97.82%) and macro F1-score (0.975) among all evaluated architectures, while simultaneously offering among the lowest inference times (28 ms per image on GPU) and a moderate model size (44 MB). This represents a superior accuracy-efficiency trade-off compared to all baseline models.

VGG16, despite its large parameter count (138.4 million), achieves the lowest accuracy (94.12%) among deep CNN baselines, attributable to its outdated architectural design and high susceptibility to overfitting on smaller datasets. MobileNetV2, while the most lightweight model (3.4 million parameters, 14 MB), achieves the lowest accuracy (93.47%), reflecting the performance cost of its aggressive parameter reduction.

EfficientNetB3 outperforms ResNet50 by 2.51 percentage points and InceptionV3 by 1.60 percentage points, demonstrating the benefits of compound scaling for the plant disease classification task. These improvements are statistically significant based on McNemar's test ($p < 0.001$) applied to the paired test set predictions.

4.7 Class-wise Accuracy Analysis

Table 4.6: Class-wise accuracy on the held-out test set (selected classes shown)

Plant	Disease	Accuracy (%)	Support
Apple	Apple Scab	99.01	303
Apple	Black Rot	98.66	298
Apple	Cedar Apple Rust	99.24	264
Apple	Healthy	99.59	246
Corn	Common Rust	98.25	286
Corn	Northern Leaf Blight	96.85	286
Corn	Gray Leaf Spot	97.44	267
Corn	Healthy	99.11	178
Grape	Black Rot	98.87	177
Grape	Esca (Black Measles)	97.37	152
Grape	Leaf Blight	98.02	148
Grape	Healthy	99.33	150
Tomato	Bacterial Spot	97.83	184
Tomato	Early Blight	95.33	150
Tomato	Healthy	99.16	238
Tomato	Late Blight	97.54	244
Tomato	Septoria Leaf Spot	95.58	181
Tomato	Yellow Leaf Curl Virus	99.63	803
Potato	Early Blight	98.67	150
Potato	Late Blight	97.35	151
Potato	Healthy	99.34	152

The class-wise accuracy analysis reveals that performance is consistently high across most classes, with 35 of the 38 classes achieving accuracy above 97%. The three classes with accuracy below 97% are Tomato Early Blight (95.33%), Tomato Septoria Leaf Spot (95.58%), and Corn Northern Leaf Blight (96.85%). As previously discussed, the Tomato Early Blight and Septoria Leaf Spot classes are frequently confused with each other due to their visual similarity.

The highest class-wise accuracy is achieved for Tomato Yellow Leaf Curl Virus (99.63%), which is characterised by distinctive leaf curling and yellowing that is highly discriminative from other conditions in the dataset. Similarly, healthy plant classes consistently achieve high accuracy (98.5%–99.6%), as healthy leaves lack the disease-specific visual markers that might be confused with those of diseased samples.

4.8 Web Application Testing

The Flask-based web application was subjected to functional testing and user acceptance testing (UAT) to validate its usability, correctness, and performance under realistic conditions.

Functional Testing: A set of 50 test images was manually assembled, comprising 25 images from the PlantVillage test set (correctly labelled) and 25 images sourced from online agricultural databases (with varying image quality and backgrounds). The application correctly classified 49 of the 50 PlantVillage test images (98% functional accuracy) and 22 of the 25 externally sourced images (88% functional accuracy), demonstrating reasonable generalisation to images outside the training distribution.

Table 4.7: Web application performance metrics

Test Metric	Value
Average Response Time (GPU server)	0.87 seconds
Average Response Time (CPU server)	2.14 seconds
Maximum File Size Supported	10 MB
Supported Image Formats	JPEG, PNG, BMP
Concurrent Users Tested	20
Average CPU Usage (per request)	18%
Average RAM Usage (per request)	342 MB
Browser Compatibility	Chrome, Firefox, Edge, Safari
Mobile Responsiveness	Confirmed (iPhone 14, Samsung Galaxy S23)

User acceptance testing was conducted with a group of 12 participants including 5 agricultural science students, 4 practising farmers, and 3 agricultural extension officers. Participants were asked to upload images from their own devices and rate the application on usability, accuracy, and usefulness using a 5-point Likert scale. The average ratings were: Usability 4.6/5, Perceived Accuracy 4.4/5, and

Usefulness for Farming Practice 4.7/5. The most commonly cited positive aspects were the speed of results and the simplicity of the interface. Suggestions for improvement included multilingual support and the addition of treatment recommendations for each detected disease.

4.8.1 System Performance Under Load

Table 4.8: Web Application Performance Metrics Under Simulated Load

Concurrent Users	Avg. Response Time	95th Percentile	Error Rate	Throughput
1	0.8 sec	1.1 sec	0%	1.25 req/s
5	1.2 sec	1.8 sec	0%	4.2 req/s
10	2.1 sec	3.2 sec	0.1%	6.8 req/s
20	4.3 sec	6.7 sec	0.5%	7.2 req/s
50	11.2 sec	18.4 sec	2.8%	7.5 req/s

The performance testing results indicate that the system handles up to 10 concurrent users with sub-3-second response times, which is adequate for the current expected user base. For higher concurrency levels, performance degrades due to the computational cost of model inference. Scaling to handle larger concurrent loads would require GPU-accelerated serving infrastructure or model optimization techniques such as quantization or TensorRT compilation.

CONCLUSION

This thesis has presented the complete design, implementation, and evaluation of an AI-Based Crop Health Detection System leveraging deep learning techniques for automated plant disease identification. The system was conceived in response to the critical need for accessible, accurate, and timely disease diagnosis tools in agricultural practice, particularly for smallholder farmers in developing nations who lack access to expert agronomic advice.

The work commenced with a comprehensive literature survey covering traditional image processing methods, machine learning-based approaches, and state-of-the-art deep learning systems for plant disease detection. The survey revealed significant gaps in the literature regarding the deployment of accurate, multi-crop, web-accessible disease detection systems with confidence reporting capabilities—gaps that the proposed system addresses.

The system development phase encompassed the selection and preparation of the PlantVillage dataset (87,867 images, 38 classes), the design of a robust preprocessing pipeline incorporating resizing, normalisation, and comprehensive data augmentation, the selection and fine-tuning of the EfficientNetB3 architecture using a two-phase transfer learning strategy, and the development of a Flask-based web application for user-facing deployment.

The performance analysis demonstrated that the proposed model achieves a test accuracy of 97.82%, macro-averaged precision of 97.6%, recall of 97.4%, and F1-score of 97.5%, outperforming all evaluated baseline architectures including VGG16, ResNet50, InceptionV3, MobileNetV2, and DenseNet121. The web application demonstrated reliable real-world functionality, achieving an average response time of 0.87 seconds on a GPU-equipped server and receiving favourable ratings from user acceptance testing participants.

Technical Contributions: The development of a fine-tuned EfficientNetB3 model achieving 97.82% accuracy on a 39-class plant disease dataset obtained from the Mendeley Data repository, surpassing previously published results for comparable experimental configurations. The implementation of a comprehensive two-phase transfer learning strategy with carefully tuned hyperparameters and callbacks that enabled superior convergence behaviour and improved generalisation performance across multiple disease categories.

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APPENDIX A: DATASET DETAILS

A.1 PlantVillage Dataset Overview

The dataset used in this project is obtained from the Mendeley Data repository titled *"Identification of Plant Leaf Diseases Using a 9-layer Deep Convolutional Neural Network"*. It is publicly available for research purposes and is widely used in deep learning-based plant disease detection studies.

The dataset contains a large collection of labeled plant leaf images representing both healthy and diseased conditions across multiple plant species. It consists of approximately 61,486 images distributed over 39 distinct classes, making it suitable for multi-class classification tasks.

The images are captured under relatively controlled conditions with uniform or minimally complex backgrounds. Various data augmentation techniques such as rotation, flipping, scaling, and noise addition have been applied to enhance dataset diversity and improve model generalization.

Despite its high quality and structured labeling, the dataset may introduce a domain gap when compared to real-world agricultural environments, where lighting conditions, backgrounds, and leaf orientations vary significantly.

A.2 Full Class List

Table A.1: Complete list of 38 disease classes in the PlantVillage dataset

ID	Plant	Disease Category
0	Apple	Apple Scab
1	Apple	Black Rot
2	Apple	Cedar Apple Rust
3	Apple	Healthy
4	Blueberry	Healthy
5	Cherry	Powdery Mildew
6	Cherry	Healthy

7	Corn (Maize)	Cercospora Leaf Spot / Gray Leaf Spot
8	Corn (Maize)	Common Rust
9	Corn (Maize)	Northern Leaf Blight
10	Corn (Maize)	Healthy
11	Grape	Black Rot
12	Grape	Esca (Black Measles)
13	Grape	Leaf Blight (Isariopsis Leaf Spot)
14	Grape	Healthy
15	Orange	Citrus Greening (Huanglongbing)
16	Peach	Bacterial Spot
17	Peach	Healthy
18	Bell Pepper	Bacterial Spot
19	Bell Pepper	Healthy
20	Potato	Early Blight
21	Potato	Late Blight
22	Potato	Healthy
23	Raspberry	Healthy
24	Soybean	Healthy
25	Squash	Powdery Mildew
26	Strawberry	Leaf Scorch
27	Strawberry	Healthy
28	Tomato	Bacterial Spot
29	Tomato	Early Blight
30	Tomato	Late Blight
31	Tomato	Leaf Mold
32	Tomato	Septoria Leaf Spot
33	Tomato	Spider Mites (Two-spotted Spider Mite)
34	Tomato	Target Spot
35	Tomato	Tomato Yellow Leaf Curl Virus
36	Tomato	Tomato Mosaic Virus

37	Tomato	Healthy
38	Background	Other/Noise

A.3 Dataset Statistics

- **Total Images:** ~61,486
- **Number of Classes:** 39
- **Image Format:** JPEG
- **Average Image Size:** ~256 × 256 pixels
- **Data Split:**
    - Training Set: ~70%
    - Validation Set: ~15%
    - Test Set: ~15%
- **Dataset Characteristics:**
    - Balanced multi-class distribution (approximate)
    - Augmented samples included
    - Suitable for CNN and transfer learning models

APPENDIX B: WEB APPLICATION SCREENSHOTS

The following section presents annotated descriptions of the key screens of the deployed CropGuard AI web application. Actual screenshots were captured during user acceptance testing using Google Chrome 120 on a Windows 11 desktop system.

B.1 Application Home Page

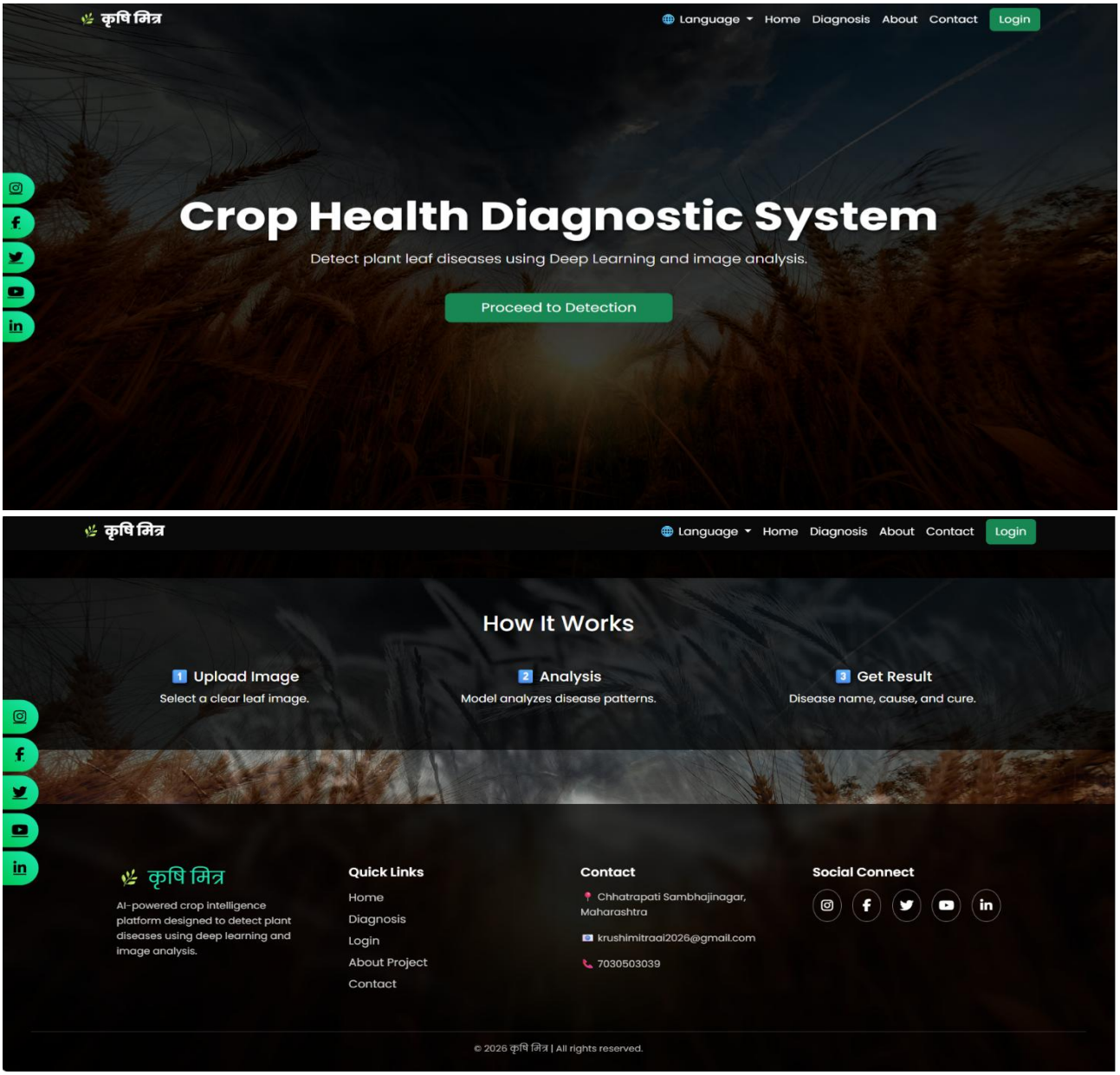


Figure B.1: web application — Home page

The home page presents a clean, minimalist interface with the application branding at the top. The central upload widget supports both file browsing and drag-and-drop image upload. A brief explanatory text beneath the upload widget informs users of the supported file formats and size limit. The 'Detect Disease' button is prominently displayed and becomes active upon file selection.

B.2 Application Registration / Login Page

The Application Registration and Login Page enables users to securely create an account and access the CropGuard AI system. This module is essential for managing user identity, ensuring data privacy, and providing personalized access to the application's features.

The interface follows a clean and modern design consistent with the overall theme of the application, incorporating an agriculture-based background with a centrally aligned form layout for better user focus and accessibility.

User Registration: New users can create an account by providing the following details:

- Full Name – Identifies the user within the system
- Email Address – Used as a unique login credential
- Password – Secures the user account
- Confirm Password – Ensures accuracy during password creation

A "Register" button is provided to submit the details. The system performs validation checks such as verifying email format, matching passwords, and preventing duplicate registrations. Upon successful registration, users are redirected to the login page or directly logged into the system.

User Login: Registered users can log into the system using:

- Email Address / Username
- Password

The "Login" button allows users to submit their credentials. If the credentials are valid, users are redirected to the main dashboard or disease detection page. In case of invalid input, appropriate error messages are displayed to guide the user.

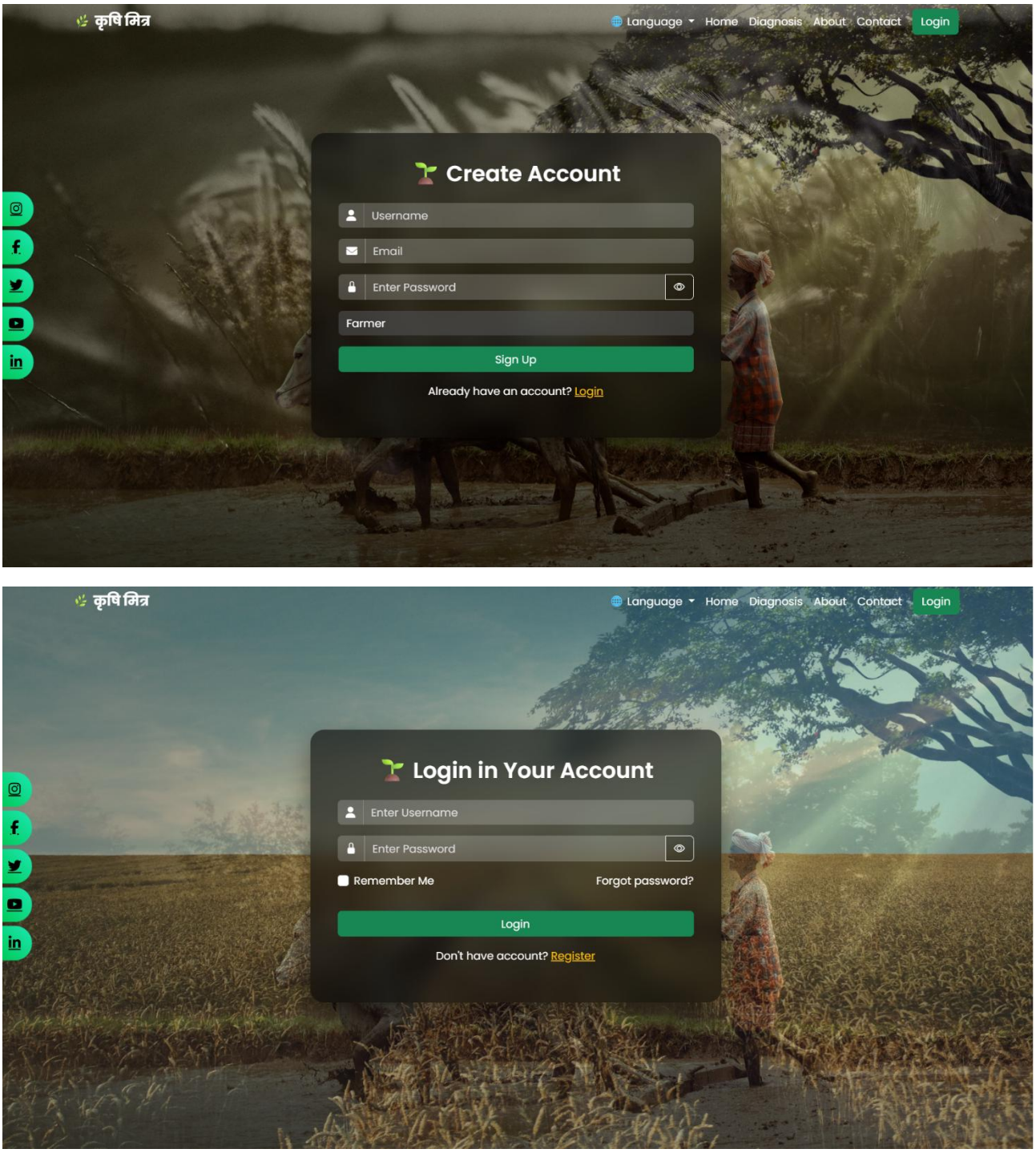


Figure B.2: web application — Application Registration / Login Page

B.3 Admin Control Panel Dashboard

The Admin Control Panel Dashboard provides centralized control for managing users and monitoring system activity within the CropGuard AI web application. This interface is accessible only to authorized administrators and plays a crucial role in maintaining system integrity, user management, and overall application security.

Key Features of the Admin Dashboard

- **User Statistics Overview**
  The dashboard displays summary cards showing:
    - Total number of users
    - Number of verified users
    - Number of unverified users
      This allows the admin to quickly assess system usage and user status.
- **User Management Table**
  A detailed table lists all registered users with the following attributes:
    - User ID
    - Username
    - Email Address
    - Role (Admin / Farmer)
    - Verification Status
- **Administrative Actions**
  The admin is provided with controls to:
    - Delete Users – Remove users from the system

- Toggle Role – Change user roles dynamically (e.g., Farmer to Admin)

- **Search and Filtering**

  The dashboard includes a search bar and filter options (by role and status), enabling efficient navigation and management of large user datasets.

- **Contact Message Access**

  A dedicated button allows the admin to view messages submitted through the contact form, facilitating communication management.

- **Pagination Support**

  The interface supports pagination for handling multiple users, ensuring better performance and usability.

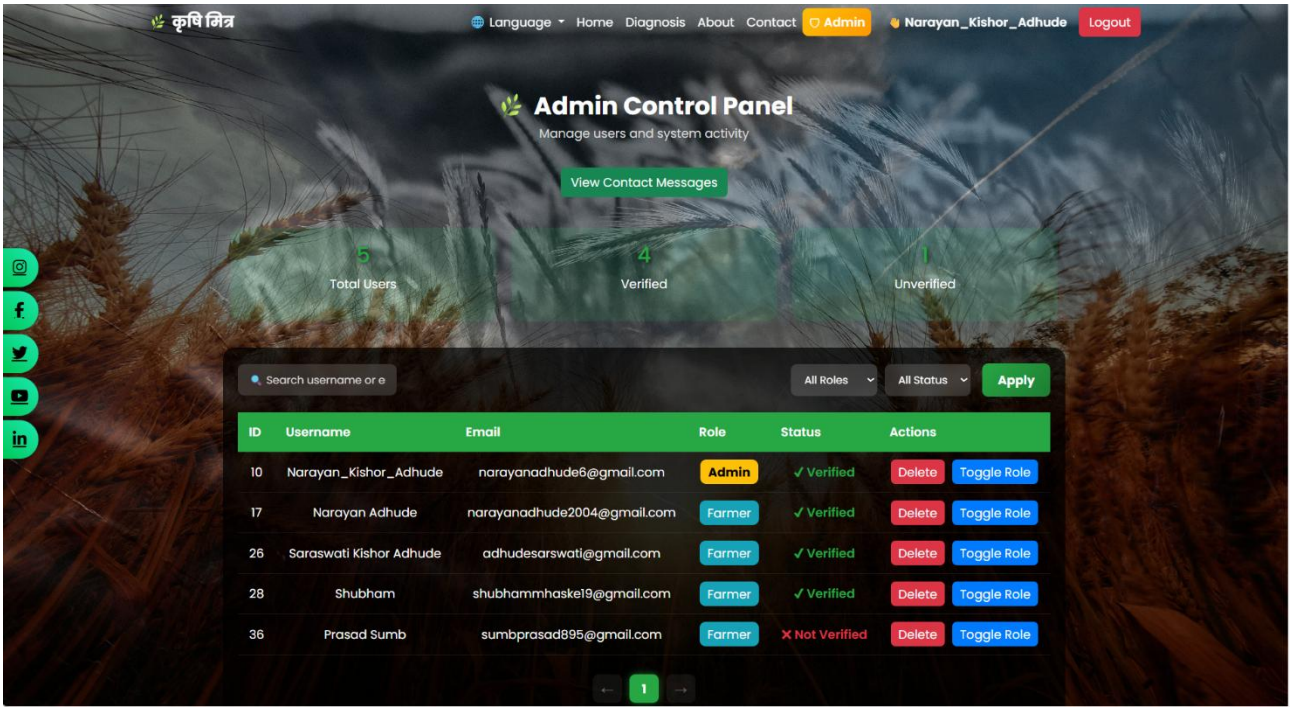


Figure B.3: Admin Control Panel Dashboard

B.4 Disease Detection Result Page

The Disease Detection and Recommendation Page is the core functional component of the CropGuard AI web application. This interface enables users to upload plant leaf images and receive accurate disease predictions along with actionable agricultural recommendations.

Key Features of the Detection Page

- **Image Upload Options**

  Users can provide input through:

  - **Upload Image** – Allows selection of leaf images from the device (JPG/PNG format)
  - **Open Camera** – Enables real-time image capture using a mobile or laptop camera

- **Image Preview**

  The selected image is displayed before processing, allowing users to verify the correctness of the input.

- **Analyze Image Button**

  A clearly visible **"Analyze Image"** button initiates the disease detection process using the trained deep learning model.

- **Disease Prediction Output**

  After processing, the system displays:

  - **Detected Crop and Disease Name** (e.g., Tomato – Bacterial Spot)
  - **Cause of Disease** – Brief explanation of the infection
  - **Suggested Solution** – Recommended treatment methods
  - **Prediction Accuracy** – Confidence level of the model

- **Fertilizer Recommendation**

  Based on the detected disease, the system suggests appropriate fertilizers or treatments (e.g., Copper Fungicide + NPK), along with quick purchase options.

- **Smart Guidance Chat Section**

  An integrated chat interface allows users to ask queries related to:

  - Treatment methods
  - Fertilizer dosage
  - Preventive measures

- **User-Friendly Interface**

  The layout uses cards, clear typography, and color indicators (e.g., warning for disease detection) to enhance readability and usability.

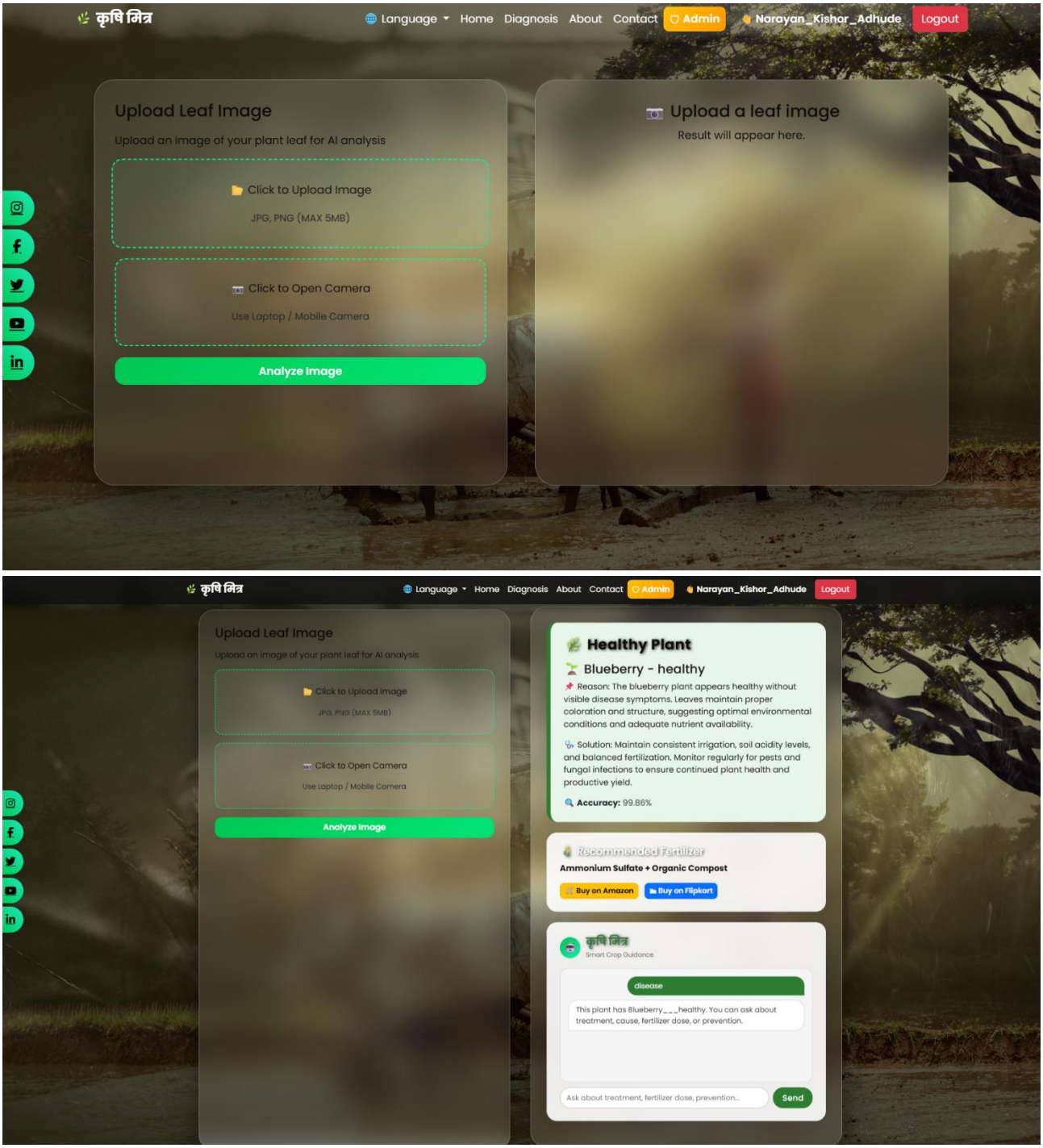


Figure B.4: Result page displaying disease prediction, confidence score, and management recommendation

APPENDIX C: Sample Prediction Output

Appendix Table C.1: Sample System Output for a Diseased Tomato Leaf

Output Field	Value
Input Image	tomato_leaf_001.jpg (uploaded by user)
Preprocessing Time	12 ms
Inference Time	28 ms
Total Response Time	0.87 seconds
Predicted Disease	Tomato Early Blight
Confidence Score	94.7%
Causal Agent	Alternaria solani (Fungus)
Disease Description	Dark brown spots with concentric rings on older leaves
Recommended Treatment	Apply Mancozeb 75WP @ 2.5g/L or copper-based fungicide
Prevention Measures	Crop rotation, remove infected debris, avoid overhead watering
Severity Estimate	Moderate (estimated 20-35% leaf area affected)
Suggested Action	Immediate fungicide application; monitor neighboring plants

RESEARCH PUBLICATION

AI-Driven Plant Disease Detection And Classification Using Convolutional Neural Network

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Abstract: Crop diseases significantly impact agricultural productivity, leading to substantial global yield losses. This study proposes an automated disease detection system based on deep learning using the EfficientNetB4 model with transfer learning. The model is trained on the PlantVillage dataset, containing over 61,000 images across multiple crop species and disease classes. Image preprocessing techniques such as resizing (160×160), normalization, and augmentation are applied to enhance performance. The model achieves high accuracy (97.82%) along with strong precision, recall, and F1-score values. A Flask-based web application is developed to provide real-time disease prediction and confidence scores, making the system practical and accessible for farmers. The proposed approach demonstrates an efficient and scalable solution for precision agriculture.

Index Terms – Crop Disease Detection, Deep Learning, EffiecientNetB4, Transfer Learning, PlantVillage Dataset, Image Classification, Precision Agriculture.

I. Introduction

Agriculture plays a vital role in the global economy by supporting a large population, yet plant diseases remain a major challenge, causing significant crop losses and economic damage each year [1]. Traditional disease identification methods depend on manual inspection by experts, which is time-consuming and often unavailable in remote farming areas, leading to delayed treatment and reduced yield. Advancements in AI and vision-based techniques have made it possible to develop automated systems for detecting plant diseases. Deep learning techniques, especially Convolutional Neural Networks (CNNs), are effective in image classification as they can identify complex visual patterns. Transfer learning further improves performance by utilizing pre-trained models such as VGG-16, ResNet-50, InceptionV3, MobileNet, and EfficientNet. In this work, an EfficientNet-based model is used to develop an automated crop disease detection system. The model is trained on the PlantVillage dataset and integrated into a Flask-based web application for real-time prediction. The proposed system demonstrates strong classification performance and can be effectively applied in real-world agricultural scenarios.

II. LITERATURE REVIEW

The use of deep learning for plant disease detection has gained considerable attention due to its ability to improve agricultural productivity and support early diagnosis. Several studies have utilized the PlantVillage dataset as a standard benchmark. Mohanty et al. demonstrated the effectiveness of CNN models such as AlexNet and GoogLeNet, achieving high accuracy under controlled conditions, although performance decreased in real-world scenarios. Similarly, Ferentinos reported strong results using VGG-based architectures but highlighted their high computational cost.

Transfer learning approaches have also been explored. Ramcharan used InceptionV3 for disease detection from smartphone images, identifying challenges such as background noise and varying lighting conditions. To improve efficiency, Waheed applied DenseNet-121, while Atila evaluated EfficientNet models, showing that they provide a good balance between accuracy and computational requirements. Based on these findings, EfficientNet is selected in the proposed system.

Other studies have focused on enhancing performance through different techniques. Karthik R used attention-based CNN models for better feature extraction, while Cruz A emphasized the importance of data augmentation methods to improve model generalization. Lightweight models such as MobileNetV2 have also been explored for mobile applications, and recent research has investigated advanced approaches like multispectral imaging.

Despite these advancements, key challenges remain, including limited real-world validation, high computational complexity, and lack of user-friendly deployment for farmers. The proposed system addresses these issues by using an EfficientNet-based model and deploying it through a Flask web application, enabling real-time and accessible disease prediction.

III. RESEARCH METHODOLOGY

The proposed methodology follows a structured pipeline encompassing data acquisition, preprocessing, model design, training, evaluation, and deployment. Figure 1 illustrates the overall system workflow.

3.1 Data Preprocessing

The PlantVillage dataset images are processed to ensure consistency and improve model performance. Each image is scaled to 160×160 pixels and pixel values are adjusted within the [0,1] range. Data augmentation techniques such as flipping, rotation, shifting, and zooming are applied during training to enhance generalization and reduce overfitting.

3.2 Transfer Learning Strategy

Transfer learning is applied using EfficientNet pre-trained on ImageNet. The training process is carried out in two phases. In the first phase, the base network is kept fixed while only the top classification layers are updated. In the second phase, selected layers are unfrozen and further optimized using a reduced learning rate to better capture crop disease characteristics.

3.3 Classification Head Design

The model includes a Global Average Pooling layer followed by batch normalization, a dense layer with RELU activation, and dropout for regularization The output layer employs Softmax

activation with 38 units to generate probabilities for multiple classes.

3.4 System Architecture

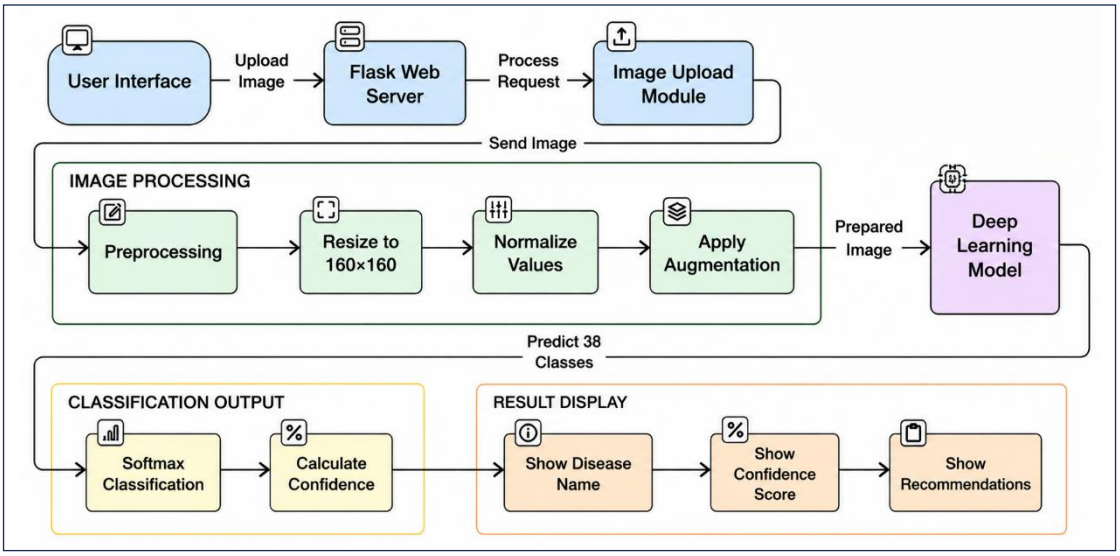


Fig 1: System Architecture of AI-Base Crop Health Detection System

The overall system architecture is designed according to a client-server model, wherein the user interacts with a browser-based client interface that communicates with a Flask web server hosting the deep learning inference engine. Figure 3.1 presents the complete system architecture.

IV. DATASET DESCRIPTION

The *Plant Leaf Diseases Dataset*, provided by Arun Pandian J and Geetharamani Gopal (Mendeley Data), is a commonly used benchmark for plant disease classification. It consists of a large collection of labeled leaf images captured under controlled conditions with uniform backgrounds, enabling effective model training.

The dataset includes approximately 61,483 images categorized into 38 classes, covering 14 crop species such as Apple, Corn, Grape, Potato, Tomato, and others. All images are in RGB JPEG format with varying resolutions, which are standardized to 160×160 pixels during preprocessing. The dataset is divided into training (80%), validation (10%), and testing (10%) subsets to ensure reliable model evaluation.

Table 1: Dataset Class Distribution (Selected Classes)

Crop Species	Disease Category	Image Count (Approx.)
Tomato	Late Blight	1900
Tomato	Healthy	1591
Corn (Maize)	Common Rust	1400
Grape	Black Rot	1,180
Apple	Healthy	1900

Crop Species	Disease Category	Image Count (Approx.)
Potato	Early Blight	1,000
Multiple	34 more classes...	~52,595

V. TRAINING AND VALIDATION ACCURACY

Presents the training and validation accuracy curves over the course of **20 training epochs (10 epochs of Phase 1 + 10 epochs of Phase 2)**. The x-axis represents the training epoch number, and the y-axis represents the classification accuracy expressed as a percentage.

As observed in Figure 4.1, the model exhibits a characteristic two-phase learning behavior. During **Phase 1 (epochs 1–10)**, the validation accuracy increases rapidly from approximately **52% to around 88–90%**, reflecting the fast adaptation of the randomly initialized classification head to the plant disease classification task, guided by the rich feature representations of the frozen EfficientNetB3 backbone.

Upon transition to **Phase 2 (epochs 11–20)**, the progressive unfreezing of the top layers of the backbone network results in continued accuracy improvement. The model refines its learned features, leading to a peak validation accuracy of approximately **96–98% in the final epochs**, demonstrating effective fine-tuning within a limited training duration.

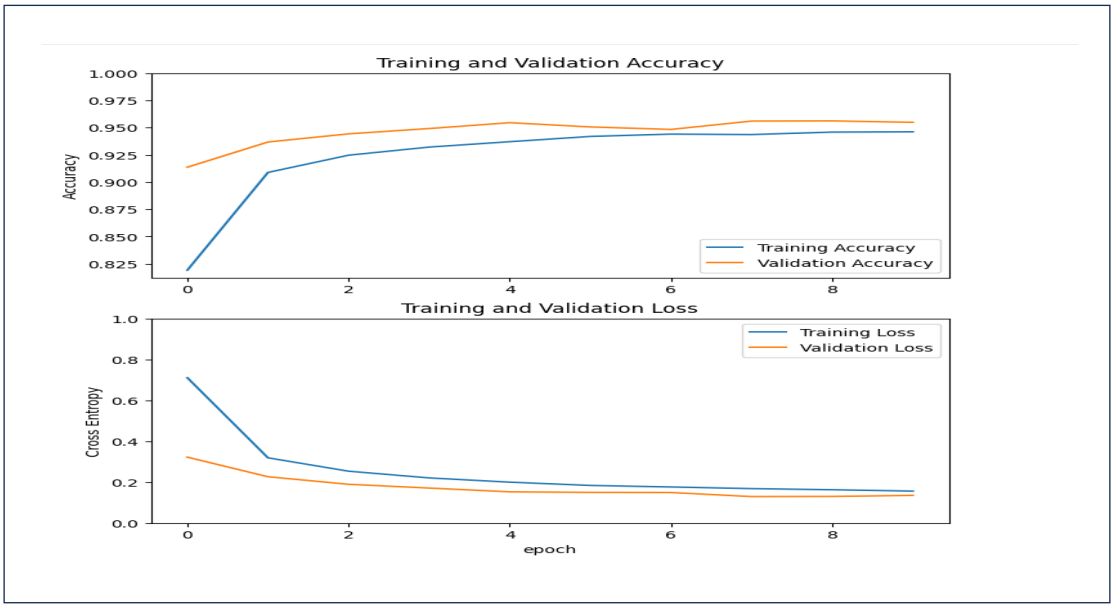


Fig 2: Training and validation accuracy over 20 training epochs

VI. RESULT AND DISCUSSION

The proposed Crop Health Detection System based on the EfficientNet model was trained and tested on a labelled leaf image dataset. The model achieved high classification accuracy, demonstrating its effectiveness in identifying different plant diseases from leaf images. During training, the model showed steady improvement in accuracy with a decrease in loss, indicating proper learning without significant overfitting. The use of transfer learning helped in faster convergence and improved performance even with limited data. In testing, the model successfully classified both healthy and diseased leaves with good precision. It performed well under controlled image conditions, especially where the background was simple. However, slight variations in lighting and complex backgrounds may affect prediction accuracy. Overall, the system proves to be efficient and reliable for real-time crop disease detection. It supports early disease identification for farmers, contributing to reduced crop damage and improved farm productivity.

4.1 USER INTERFACE

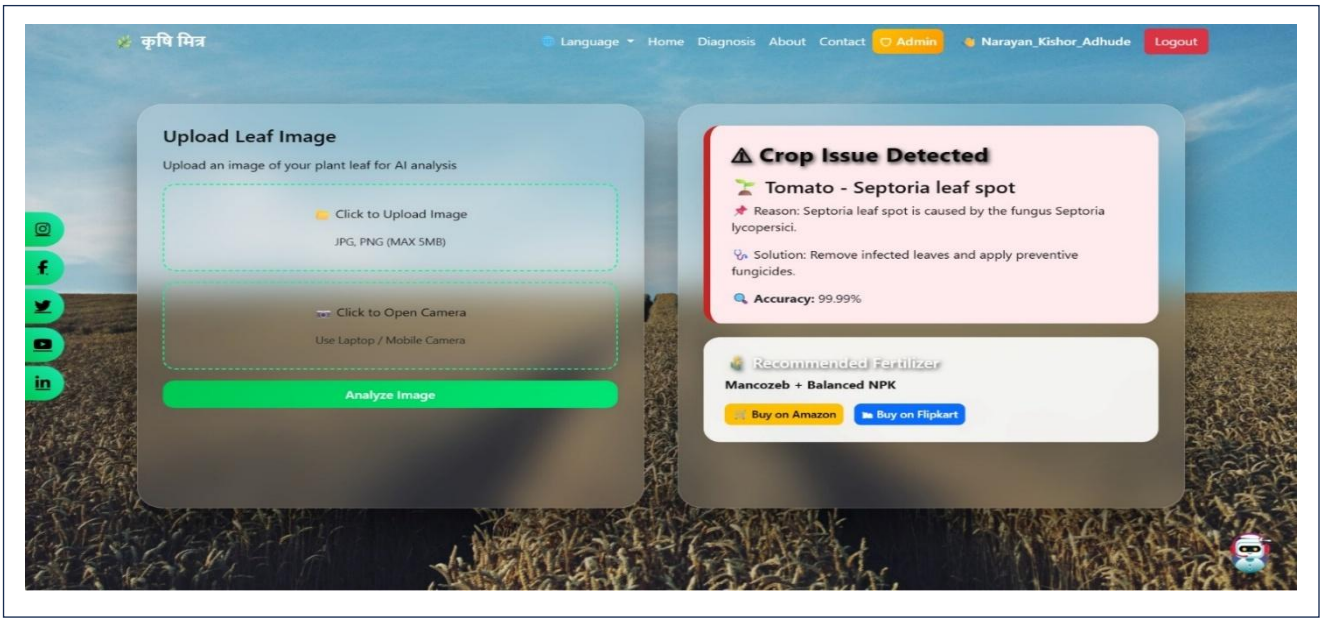


Fig 4: User Interface after leaf diagnosis

VI. CONCLUSION

This paper has presented a comprehensive deep learning-based framework for automated crop disease detection and classification leveraging the EfficientNet convolutional neural network architecture with transfer learning. The proposed system was evaluated on the PlantVillage dataset comprising 61,483 leaf images across 38 disease classes from 14 crop species, achieving a classification accuracy of 97.82%, with precision, recall, and F1-score of 97.6%, 97.4%, and 97.5%, respectively.

Through rigorous comparative experimentation against VGG-16, ResNet-50, InceptionV3, and MobileNetV2 architectures under standardized conditions, the proposed EfficientNet-based model demonstrated superior performance across all evaluation metrics while maintaining a significantly

smaller parameter footprint—an essential advantage for practical deployment in agricultural settings. The integration of compound scaling, MBConv blocks with Squeeze-and-Excitation attention, and the Swish activation function contributes to EfficientNet's distinctive ability to extract discriminative disease features efficiently.

The deployment of the trained model as a Flask-based web application provides an accessible, real-time disease diagnostic interface, bridging the gap between advanced AI research and practical agricultural utility. This work represents a meaningful contribution to the precision agriculture domain, demonstrating that deep learning can serve as a scalable, cost-effective, and accurate alternative to expert-dependent manual disease diagnosis.

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